

The Invisible Hand of Internal Markets in Mutual Fund Families ^{*}

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Abstract

The internal markets of fund families can encourage member funds to deviate excessively from their investment mandates. Theoretically, we show that fund managers following sufficiently different style benchmarks can engage in risk-shifting by trading with one another at low cost inside their family. This benefits the managers and the family even in the absence of a family-level strategy. However, the excessive risks taken by the managers can be costly to fund investors. Empirically, we find support for the positive effect of intra-family style diversity on offsetting trades across funds and on deviations of funds' portfolios from their benchmarks.

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Abstract

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1 Introduction

Mutual fund families can potentially make up very large internal markets. The median family manages nearly \$6 billion in assets and has 5 funds under its umbrella. Moreover, up to 35% of the portfolios of funds affiliated with the same family is invested in the same stocks. Funds in need of rebalancing their portfolios will thus likely search within their family first for counterparties to their trades, as trading in the internal market is less costly compared to trading with the external market.¹ In this paper, we argue both theoretically and empirically that the internal markets within fund families not only reduce transaction costs for their member funds but also increase the potential for misalignment between the investment policies of the funds and the mandates of their respective shareholders.

We model a family consisting of two funds that follow different benchmarks. Fund managers maximize expected utility of end-of-period compensation, which is positively related to the money flows of the fund investors. Building on the findings of Sirri and Tufano (1998) and Basak, Pavlova, and Shapiro (2007), we assume that investors' flows are a convex function of funds' past performance relative to the performance of their style benchmarks (DelGuercio and Tkac (2002)). In order to motivate trading in the internal market, we account for the empirically documented overlap in stock holdings inside a family by assuming that member funds share some of their portfolio holdings. In the model, the internal market offers an advantage over the external market for trades in illiquid holdings. We then allow the two fund managers in our family to meet at the start of the investment period and decide whether it is in their mutual best interest to cross-trade some of their illiquid holdings. After this initial meeting, managers choose their funds' investment policies independently of each other.

The first outcome of our model is that style diversity alone can induce cross-trading

¹ Chalmers, Edelen, and Kadlec (2013) estimate that the annual trading costs for equity funds are of first order relevance, averaging 1.44% of fund assets. In line with these estimates, a survey study conducted by the Bank for International Settlement (BIS (2003)) points to savings on transaction arising from "crossing of trades" as one of the main factors behind the trend towards consolidation among investment managers in the asset management industry.

within a fund family. Trades in the same asset but in opposite directions can happen even when fund managers share the same information and have the same preferences. Moreover, the cross-trading in our model is not coordinated at the family-level but results from the optimal decentralized investment decision of the individual fund managers. The mechanism is as follows: (i) fund managers seek to beat their benchmarks to increase their personal compensation; (ii) to improve the chance of beating their benchmarks, managers need to deviate their portfolio composition away from the composition of their benchmarks; (iii) a manager following a risky benchmark deviates by taking safe bets, while a manager following a safe benchmark deviates by taking risky bets; (iv) when deviating at the same time, the first manager could be underweighting a risky illiquid asset that the second manager seeks to overweight in her portfolio, which creates the opportunity for cross-trading. The internal market within the mutual fund family facilitates these interfund trades, even without the need for a family-level strategy.

We show that the more dispersed the styles that these funds follow, the higher the probability that they place opposite orders on the same asset, and the larger the expected magnitude of the internal trade. As a result, cross-trading increases monotonically with style dispersion and can be substantial in diverse enough families. Style diversity is then both a necessary condition for, and positively related to, the decentralized cross-trading in our model. In the empirical section of the paper we find support for this prediction.

Throughout the investment period, fund managers do not necessarily rebalance their portfolios towards their respective benchmark compositions. Hence, the initial cross-trade along with the subsequent trading may alter the average portfolio liquidity of the family-affiliated funds relative to their benchmarks in a way that differs significantly from standalone funds (i.e., equivalent funds with no possibility of cross-trading). When following a low-liquidity style, funds increase liquidity relative to the benchmark more when they belong to a family. Conversely, funds following a high-liquidity style decrease portfolio liquidity more

when affiliated with a family. Moreover, the deviation of funds' liquidity from the liquidity of their benchmarks increases with style diversity within the family.

As if led by an “invisible hand” of internal markets, the decentralized cross-trading that maximizes each of the fund manager's utility also increases the benefits accruing to the family as a whole despite the absence of a family-level strategy.² It helps increase the value of future assets under management for the family, because it allows fund managers to take advantage of the relation between performance and future flows to a larger extent. We show that the benefits accruing to the family increase with the dispersion of styles within the family. These results provide an alternative rationale for fund families to offer a diverse menu of investment styles across their funds, as observed in practice.

There are two main drivers of cross-trading in our model. First, as the intra-family style diversity increases, benchmarks become more concentrated, and funds are encouraged to deviate from their benchmarks to diversify their portfolios. This is the case even in the absence of convex incentives. Second, funds deviate from their benchmarks to increase the likelihood of future flows, which are a convex function of the funds' benchmark-adjusted performance. In order to distinguish these two motives in the model, we remove the convexity in the flow-performance relation. We show that the risk-shifting incentive to deviate from benchmarks can be costly. The resulting costs of cross-trading are borne by the investors of at least one of the funds, who would be better off by delegating to a standalone fund. Thus, investors' decision to invest in family-affiliated versus standalone funds involves trading off the benefit of lower transaction costs versus the higher risk-shifting costs that result from the availability of a cross-trading platform.

We use a sample of U.S. domestic actively managed equity mutual funds to examine empirically the novel predictions of our model. We test the hypothesis that the style diversity

² The term “invisible hand” was introduced by Adam Smith to describe his belief that individuals seeking their economic self-interest actually benefit society more than they would if they tried to benefit society directly.

offered by a family is positively related to (i) the level of intra-family offsetting trades, and (ii) the deviation of funds' portfolio liquidity with respect to the liquidity of their style benchmarks. We also examine the implications of cross-trading for fund performance. Our prior is that, in the industry equilibrium, funds should engage in cross-trading until its benefits and costs exactly offset.

We measure intra-family style diversity through the correlations of returns across all the benchmarks followed by the funds affiliated with the same family.³ We find that the diversity in styles offered by a mutual fund family is a first-order determinant of the offsetting trades within the family. Even after controlling for strategic cross-fund subsidization (Gaspar, Massa, and Matos (2006)), the extent of overlap in holdings, and the flow correlations across funds affiliated with the same family, a one standard deviation increase in style diversity is associated with a 21% increase in intra-family offsetting trades across all stocks. The figure raises to 25% when we focus on offsetting trades of illiquid stocks only. As predicted by our model, this effect should only be present *within* families. We show that this is indeed the case, as style diversity has no effect on the level of interfund trading for placebo families, which we create by randomly drawing for each of the actual families comparable funds from all the other families in our sample.

We also find that style diversity is positively associated with the deviation in liquidity between funds' portfolios and their style benchmarks. As predicted by our model, funds following high-illiquidity benchmarks deviate by decreasing the illiquidity of their portfolios, and vice-versa for funds following low-illiquidity benchmarks. We show that these portfolio distortions are unrelated to the cross-subsidization motive of Gaspar, Massa, and Matos (2006).

³ Our measure of intra-family style diversity is defined as the (negative of the) minimum of all the pairwise correlations of returns across the different benchmarks followed by the funds affiliated with the same family. The composition of these benchmark portfolios is outside of the control of the fund family. This way we avoid the endogeneity issues that would arise if pairwise correlations were computed using fund returns instead of benchmark returns.

Regarding the implications of intra-family style diversity on fund performance, our model suggests that the net effect should be positive for passive funds, but zero for active funds. This is because passive funds only engage in cross-trading to save in trading costs, while active funds engage in cross-trading not only to save in trading costs, but also for risk-shifting purposes. Our empirical evidence is also consistent with these predictions.

Our paper is related to the growing literature on cross-trading within asset management companies. In their seminal work, Gaspar, Massa, and Matos (2006) find that one way in which mutual fund families can transfer performance across member funds ('cross-fund subsidization'), to favor those funds with a higher expected contribution to family profits, is to have them cross-trade at below or above market prices. Chaudhuri, Ivkovic, and Trzcinka (2017) find evidence of a similar strategic performance re-allocation across the different products offered by an institutional asset management company, with stronger effects occurring within illiquid investment styles. Casavecchia and Tiwari (2016) document a similar effect for brokers and other clients of the fund advisor, at the expense of the mutual fund clients. Eisele, Nefedova, Parise, and Peijnenburg (2017) provide transaction-level evidence of the cross-subsidization incentive hypothesized in Gaspar, Massa, and Matos (2006).⁴

We contribute to this literature in at least two ways. First, existing studies emphasize centralized decisions that favor some funds at the expense of others, whereas we study cross-trading that can happen without a centralized decision maker. Since actual mutual funds' trades are likely the outcome of both centralized and decentralized decisions, our results suggest that the exclusive focus on the centralized decisions of fund families may understate the real costs to fund investors derived from interfund transactions. Second, most of this literature emphasizes agency costs stemming from cross-trades executed at 'transfer prices'

⁴ Other empirical studies documenting family-level strategies within mutual fund complexes include Nanda, Wang, and Zheng (2004), Bhattacharya, Lee, and Pool (2013), and Goncalves-Pinto and Schmidt (2013).

that deviate from the prevailing market prices. In the U.S. context, these trades are clearly against SEC Rule 17(a)-7 of the Investment Act of 1940, which allows cross-trading as long as the transaction is effected at the “independent current market price of the securities” and is in the best interests of *both* the selling and the buying funds. The cross-trades in our model are executed at fair market prices, and can be particularly costly to fund investors when coordinated at the family level. To the extent that investors in one or both funds are made worse off, these cross-trades still go against Rule 17(a)-7 but should arguably be harder to detect by regulatory authorities than in the cross-subsidization case.

The rest of the paper proceeds as follows. We describe our model in Section 2. We solve the model and discuss its main implications in Section 3. We discuss additional issues related to the model in Section 4. We provide empirical evidence consistent with the novel predictions of our model in Section 5. We conclude in Section 6.

2 Model Setup

2.1 The economy

We consider an economy with three dates (i.e., two periods): (1) the beginning of the calendar year ($t = 0$), the middle of the year ($t = 1$), and the year-end ($t = 2$). The financial market consists of three risky assets $i \in \{1, 2, C\}$ with prices denoted by $S_i(t)$. Initial prices for these assets are $S_i(0) = s_i$. For simplicity, we assume that each asset’s price evolves according to a two-period binomial tree.

Assets 1 and 2 trade in perfectly liquid markets, whereas asset C trades in an imperfectly liquid (‘illiquid’) market. Specifically, we follow the proportional transaction cost literature (e.g., Constantinides (1986)) in assuming that trading (i.e., buying or selling) N shares of asset C at time t incurs a transaction cost of $\theta|N|S_C(t)$.

We consider two types of investors: delegating households and fund managers. They

have constant relative risk aversion (CRRA) preferences over consumption of wealth at the terminal date $t = 2$:

$$U_k(w) = \frac{w^{1-\gamma_k}}{1-\gamma_k}, \quad (1)$$

where $\gamma_k > 0$ and $\gamma_k \neq 1$, and $k \in \{m, h\}$ denotes a fund manager m or a household h .

2.2 Investment styles and mutual funds

We assume that households are heterogeneous in risk appetite, and such heterogeneity translates into preferences for funds with different risk-return profiles, or ‘investment styles.’ Investment styles are identified by benchmark portfolios that invest in different sets of assets, not necessarily mutually exclusive. We consider two investment styles, L and S. Style L can be thought of as a conservative strategy focusing on liquid assets, such as well-known publicly traded large-cap (“L”) stocks, while style S can be thought of as an aggressive strategy investing predominantly in illiquid assets, such as small-cap (“S”) stocks. Therefore, we characterize style L by a portfolio of assets 1 and C , with a higher weight on the liquid asset 1, while we characterize style S by a portfolio of assets 2 and C , with predominance of the illiquid asset C . Thus, only the illiquid asset C is held in common by both benchmark portfolios.⁵

The style benchmark $Y_j(t)$ for $j \in \{1, 2\}$ is a passive buy-and-hold portfolio, holding B_C^j shares of asset C and B_j shares of asset j . We set $\beta_C^j = B_C^j s_C / y_j$ and assume that $0 \leq \beta_C^1 < \beta_C^2 \leq 1$ so that benchmark 1 represents the liquid style L, while benchmark 2 represents the illiquid style S. The price dynamics of benchmark j is given by:

$$Y_j(t) = B_C^j S_C(t) + B_j S_j(t) \quad (2)$$

⁵ This is without loss of generality. The introduction of a liquid asset held in common by both benchmark portfolios would not lead to cross-trading in our setup. As a result, it would not qualitatively affect our results.

with initial value $Y_j(0) = y_j$, $j \in \{1, 2\}$. We assume that benchmark j is catered to a household with RRA coefficient γ_{hj} in the sense that the weight of asset C in benchmark j (β_C^j) is the one that maximizes the utility of household h_j among all possible buy-and-hold portfolios. When asset C is riskier than assets 1 or 2, this weight decreases monotonically with the coefficient of relative risk aversion.

We consider two mutual funds denoted by fund 1 and fund 2. Fund 1 follows investment style L and is administered by portfolio manager $m1$, whereas fund 2 follows style S and is administered by portfolio manager $m2$. Let $(N_j(t), N_C^j(t))$ denote the number of shares of assets j and C that fund $j \in \{1, 2\}$ holds in the portfolio as of time t . The value of fund j 's self-financing portfolio, $F_j(t)$, evolves according to the following:

$$F_j(t) = N_C^j(t)S_C(t) + N_j(t)S_j(t) \quad (3)$$

with initial value $F_j(0) = f_j$. In addition, we assume that fund j 's initial portfolio composition is the same as that of its benchmark, i.e. $n_C^j = B_C^j$ and $f_j = y_j$.⁶ Table 1 summarizes the main elements of our model.

[Insert Table 1 about here]

Families' profits are typically a direct function of fees charged and assets managed. Therefore, fund companies are concerned not only with maximizing fund returns (as are fund shareholders), but also with attracting more assets to manage. We capture this potential incentive misalignment between a fund company and its shareholders (Gaspar, Massa, and Matos (2006)) by allowing, at $t = 2$ (also denoted as terminal date T), for external cash

⁶ Any other choice of n_C^j would arbitrarily impose a positive or negative benchmark-adjusted return on the fund portfolios right from the start of the investment period. Our assumption that the initial portfolio of the fund exactly matches its benchmark tries to circumvent this potential problem. We find that our results do not depend on this particular choice of initial portfolios for our mutual funds.

inflows to, or outflows from, the funds. Following the stylized fact documented in the literature (e.g., Sirri and Tufano (1998)), we assume that investors' flows penalize very poor and moderately poor relative performance at similar rates but reward good performance at an increasingly higher rate. Specifically, we borrow from Basak, Pavlova, and Shapiro (2007) the following flow-performance relationship:

$$\phi(R_j^F(T) - R_j^Y(T)) = \begin{cases} \phi^L & \text{if } R_j^F(T) - R_j^Y(T) < \eta \\ \phi^L + \psi \left[e^{R_j^F(T) - R_j^Y(T)} - e^\eta \right] & \text{if } R_j^F(T) - R_j^Y(T) \geq \eta, \end{cases} \quad (4)$$

where $R_j^F(T) = \ln(F_j(T)/F_j(0))$ denotes the continuously-compounded return of fund j 's portfolio, and $R_j^Y(T) = \ln(Y_j(T)/Y_j(0))$ denotes the return of benchmark j 's portfolio, ϕ^L and ψ are positive constants, and $\eta \in \mathbb{R}$. According to the flow function (4), fund j experiences outflows at a constant rate ϕ^L when its performance relative to the benchmark is lower than η , and experiences inflows at an increasing rate $\psi(e^{R_j^F(T) - R_j^Y(T)} - e^\eta)$ when its relative performance is greater or equal to η .

The flow function (4) can alternatively be interpreted as an explicit compensation contract according to which the fund family remunerates a fund manager with a fixed fee over assets under management plus a bonus conditional on meeting a given performance target. Such compensation structure has been shown to be optimal in delegated asset management contexts (e.g. Maug and Naik (2011), Basak, Pavlova, and Shapiro (2008), Li and Tiwari (2009) and Dybvig, Farnsworth, and Carpenter (2010)). Moreover, they are standard in practice in the mutual fund industry as documented by Ma, Tang, and Gomez (2017).

2.3 Family-affiliated versus standalone funds

In our model, mutual funds 1 and 2 can be part of the same fund family (FA funds) or they can be standalone entities (SA funds).⁷ In our setup, what makes a family different from just a portfolio of two SA funds is the possibility that affiliated funds can avoid public markets at $t = 0$ by trading shares of their common asset C with one another. SEC Rule 17(a)-7 of the Investment Act of 1940 allows cross-trading as long as the transactions are effected at the “independent current market price of the securities” and are in the best interest of *both* the selling and the buying funds.⁸ While we acknowledge that in practice cross-trading can happen at any of the three dates in our 2-period model (i.e. $t \in \{0, 1, 2\}$), we restrict it to occur only at the start of the year (i.e. $t = 0$) for tractability. Therefore, we interpret our results as a lower bound on the effects that can be expected in practice.

Let X be the amount of cross-trading in asset C : a positive value of X is interpreted as a purchase of asset C by fund 1 from fund 2, whereas a negative value represents a sale from fund 1 to fund 2. After cross-trading, the funds’ initial holdings of asset C change as follows:

$$N_C^1(0) = n_C^1 + X \quad \text{and} \quad N_C^2(0) = n_C^2 - X \quad (5)$$

The initial holdings of liquid assets 1 and 2 also change because the funds’ portfolios are self-financing. They change in the following way,

$$N_1(0) = \frac{n_1 S_1(0) - X S_C(0)}{S_1(0)} \quad \text{and} \quad N_2(0) = \frac{n_2 S_2(0) + X S_C(0)}{S_2(0)} \quad (6)$$

We impose the condition that $-n_C^1 \leq X \leq n_C^2$ (i.e. no ‘internal’ short-selling). If $X = 0$,

⁷ The prevalent organizational form in the U.S. mutual fund industry is the fund family; see Gaspar, Massa, and Matos (2006). We abstract from looking into the reasons behind the emergence of families as an organizational form in the first place, or the factors determining families’ optimal size, and take them as given.

⁸ The latter condition was stipulated in a no-action letter (Federated Municipal Funds (November 20, 2006)) in which the SEC staff provided clarification on Rule 17(a)-7 with respect to an investment adviser’s fiduciary duties in connection with Rule 17(a)-7 transactions.

then FA funds are exactly equal to SA funds.

After cross-trading at time 0, each FA fund is managed independently over the rest of year. Moreover, we assume that after cross-trading, funds can also trade asset C in the open market at time $t = 0$ by incurring a transaction cost. Thus, after trading in the open market, the number of shares of illiquid asset C held by each fund at time $t = 0$, is given by

$$N_C^1(0+) = N_C^1(0) + Z_1(0) \quad \text{and} \quad N_C^2(0+) = N_C^2(0) + Z_2(0) \quad (7)$$

where $Z_j(0)$ is the number of shares of asset C purchased (if positive) or sold (if negative) by fund j in the open market at $t = 0$. By the self-financing condition, the number of shares of liquid asset j in fund j ' portfolio changes as follows,

$$N_j(0+) = \frac{N_j(0)S_j(0) - Z_j(0)S_C(0) - \theta|Z_j(0)|S_C(0)}{S_j(0)} \quad (8)$$

for $j = 1, 2$.

At time $t = 1$ (mid-year), the funds can only trade asset C in the open market. They choose the number of shares of asset C to trade at $t = 1$ (i.e. $Z_j(1)$), after which the new holdings of illiquid asset C ($N_C^j(1)$) and liquid assets j ($N_j(1)$), change in a similar fashion as in equations (7) and (8), which we omit for brevity.

2.4 Fund managers' investment problem

Fund managers' objective is to maximize compensation at $t = 2$, which is a function of the funds' assets under management, in turn being a function of the inflows and outflows that funds experience at $t = 2$. Therefore, for a given level of cross-trading X at time $t = 0$, the

manager of fund j solves the following problem:

$$V_j(F_j, Y_j, N_C^j, S_C) = \sup_{Z_j(0), Z_j(1)} E \left[\frac{[\phi(R_j^F(T) - R_j^Y(T))F_j(T)]^{1-\gamma_m}}{1-\gamma_m} \right], \quad (9)$$

subject to the price processes of liquid assets 1 and 2, and of illiquid asset C , the dynamics of the assets under management in equation (3), the benchmark portfolios in expression (2), and the initial holdings given by equation (5).

In a rational framework, managers solving problem (9) are willing to engage in cross-trading only if by doing so they can increase their indirect utility, compared to the no-cross-trading case. Therefore, if we denote as $\Delta V_j(X)$ the difference in indirect utility before and after cross-trading at $t = 0$, then we assume that managers agree at the outset on the following constrained max-min rule when determining their optimal cross-trading:

$$X = \arg \max_{X'} \left(\min\{\Delta V_1(X'), \Delta V_2(X')\} \right) \quad s.t. \quad \min\{\Delta V_1(X'), \Delta V_2(X')\} > 0 \quad (10)$$

According to this rule, FA fund managers will trade with each other only if *both* strictly benefit from doing so. Moreover, the extent of cross-trading X is constrained by the maximum improvement in utility achieved by the manager that benefits the least.⁹ Note that problem (10) does not necessarily have a solution, in which case managers will not trade with each other and will behave as SA funds throughout the entire investment period.

2.5 Discussion of model assumptions

Two of the main assumptions in our model are (i) asset C is illiquid, and (ii) funds experience flows that are a convex function of the benchmark-adjusted fund performance. In this subsection we provide a discussion of these main assumptions.

⁹ We exclude the possibility of side payments between managers that could encourage otherwise suboptimal participation in cross-trading.

First, the illiquidity of asset C is a necessary condition for cross-trading to be relevant. This means that asset C could be either a small-cap stock or a mid-cap stock, as long as the stock exhibits a certain degree of illiquidity. If asset C were perfectly liquid, fund managers would have no need to cross their opposite trades within their family group. Indeed, they could trade at zero cost in the open market. However, when trading this asset in the open market is costly, under SEC Rule 17(a)-7, fund managers can reduce these trading costs by ‘crossing’ their trades within the family.¹⁰

Second, the assumed flow-performance relationship, while not strictly necessary to generate cross-trading, creates significant risk-shifting incentives (Basak, Pavlova, and Shapiro (2007)). These incentives allow us to assess the effect of cross-trading on delegation costs to a larger extent. We leave the mechanism behind the convexity of the flow-performance relation, i.e. the performance-chasing behaviour of households, outside of our model. However, the model could be extended to allow for managerial skill to potentially (holding all else constant) create value for households. This value creation process could justify the performance-chasing behaviour that we are now taking as exogenous. To the extent that these benefits are orthogonal to the costs induced by family affiliation, the conclusions of our model would remain qualitatively unchanged.

3 Model Results

3.1 Parameter values

The model defined by (1)-(10) has no analytical solution. However, we can obtain a numerical solution, for a given set of parameter values, in a rather straightforward way. To parameterize the model, we seek to match observed estimates in actual financial markets where available,

¹⁰ We assume that managers do not compete with each other within the fund family. See Taylor (2003) and Basak and Makarov (2014) for an analysis of the risk-shifting incentives induced by the strategic interaction among managers of the same family when their compensation is based on tournaments.

and borrow parameter values from previous studies otherwise.¹¹

We set the values of the parameters so that the annualized mean return and volatility of each of the three assets in each period are $\mu_1 = \mu_2 = 0.09$, $\mu_C = 0.165$, and $\sigma_1 = \sigma_2 = 0.2$, $\sigma_C = 0.3$. These parameter values imply a higher reward-to-risk ratio μ/σ for the illiquid asset C relative to the liquid assets 1 and 2, which we assume as a non-negative premium for illiquidity. Without loss of generality, we normalize the asset prices, the values of the funds' assets under management, and of their corresponding benchmarks, to one unit at $t = 0$, i.e., $s_C = s_j = f_j = y_j = 1$, for $j \in \{1, 2\}$. We set the initial value of the total holdings of illiquid asset C within the family $(n_C^1 + n_C^2)s_C$ to one unit as well, so that $\beta_C^2(0) = 1 - \beta_C^1(0)$.

We follow Basak, Pavlova, and Shapiro (2007) in calibrating the flow-performance relationship to approximately match the estimates in Sirri and Tufano (1998), by setting $\phi^L = 0.97$, $\psi = 1.6$, and $\eta = -0.05$. We set the fund managers' risk-aversion coefficients to $\gamma_m = 3.5$, in line with the estimates of managerial risk aversion in Kojien (2014).

As discussed in subsection 2.2, the “style diversity” between the two benchmarks is a function of the coefficient of RRA of household $h1$ (i.e. γ_{h1}) to whom style L is catered. We vary γ_{h1} between 6 and 9 to consider the implications of various “style diversity” scenarios. Therefore, as in subsection 2.2, this maps into an initial weight in the illiquid asset C for benchmark 1, i.e. $\beta_C^1 = \beta_{unc,C}^1(\gamma_{h1})$, that varies between 0.42 and 0.37. By construction, the initial weight of asset C for benchmark 2 (i.e. $\beta_C^2 = \beta_{unc,C}^2(\gamma_{h2})$) varies between 0.58 and 0.63, which corresponds to coefficients of RRA for household $h2$ (i.e. γ_{h2}), to whom style S is catered, that vary between 2.6 and 2.2.

¹¹ As with any numerical analysis, the implications we describe in this paper need not (and will generally not) hold for all parameter configurations. Therefore, they should be interpreted as holding under the set of parameter values that we are currently using. Nevertheless, the qualitative results of our model would not change significantly for alternative combinations of parameter values, provided that the cross-trading platform continues to be a valuable tool for the fund managers.

3.2 Benchmark-induced trading

A risk-averse manager can choose to deviate from its benchmark for two main reasons. First, if the manager's portfolio is underdiversified, the manager can deviate to achieve greater diversification benefits. Second, in the presence of a convex flow-performance relationship like (4), Basak, Pavlova, and Shapiro (2007) and Chen and Pennacchi (2009) argue that the manager can further deviate from the benchmark in order to increase the likelihood of future fund inflows. These deviations can vary over time, and can be achieved by overweighting or underweighting a given asset in the fund's portfolio, relative to its corresponding weight in the benchmark. The sign of the deviation will depend on the riskiness of the benchmark.¹² In practice, a typical fund family offers multiple investment styles with varying degrees of riskiness. The response of the family-affiliated fund managers to these benchmarks can then create opportunities for cross-trading, whereby managers simultaneously deviate from their benchmarks in opposite directions.

In order to examine this possibility, we solve our two-period model for the case of two SA funds (i.e. funds that have no access to a cross-trading platform). Recall that uncertainty in our model is governed in each period by a binomial tree representation, which allows us to easily extract simple statistics along the tree branches and nodes. If we assume that the weight of illiquid asset C in fund 1 is $\beta_C^1 = 0.4$ and the transaction costs rate is $\theta = 0.005$, we obtain the following outputs from our 2-period model. At $t = 1$, the probability that managers $m1$ and $m2$ trade in opposite directions is 25%, with a conditional offsetting trading volume (the minimum of buy and sell offsetting orders scaled by total trading) of 20%. In anticipation of their expected state-by-state trading at $t = 1$, the SA fund managers' offsetting trading volume at $t = 0$ is 73.7%, with manager $m1$ increasing and manager $m2$

¹² In line with this argument, Huang, Sialm, and Zhang (2011) find empirically that active mutual fund managers change their portfolio risk dynamically over time, increasing or decreasing it depending on their interim position relative to peers. Dai, Goncalves-Pinto, and Xu (2017) show that these effects weaken in the presence of liquidity restrictions, although managers' deviations from the benchmark can still remain significant.

decreasing their allocations in illiquid asset C .

These results illustrate how, at a particular point in time and in response to different benchmarks, SA funds can choose to trade the illiquid asset C in opposite directions. It is important to highlight that both managers share the same information and preferences in our model.¹³ Therefore, their tendency to trade in opposite directions is only the result of their optimal response to the benchmarks and the flow-performance relationship. The mechanism is as follows. Initially, the liquid benchmark 1 underexposes manager $m1$ to the illiquid asset C , to which $m1$ responds by increasing asset C in the portfolio relative to the benchmark. The opposite is true for manager $m2$, who is initially overexposed to the illiquid asset C and chooses to reduce the weight in the portfolio relative to that in the benchmark. As each fund's performance relative to the benchmark varies over time, managers hedge against changes in expected flows by increasing or decreasing their excess exposure in asset C .

In the following subsection, we allow these funds to have access to a cross-trading platform within the fund family, and examine the extent to which they make use of it.

3.3 Optimal cross-trading

Figure 1 illustrates how benchmark concerns can lead to substantial decentralized cross-trading between FA funds in our model. The vertical line identified as NCT (which stands for no-cross-trading) in Figure 1 represents the case of SA funds. These funds cannot access the cross-trading platform offered by a fund family, in which case $X = 0$. To the left (respectively, right) of NCT, fund 2 purchases (sells) asset C from (to) fund 1 in the internal market, in which case $X < 0$ ($X > 0$). The figure shows that both fund managers would decrease their indirect utility if fund 2 were to buy additional shares of asset C from fund 1

¹³ The symmetry in managers' risk preferences and their complete information about economic fundamentals imply that their portfolio decisions in perfectly liquid markets would be identical at every point in time. As a result, no cross-trading would occur if all assets were perfectly liquid.

in the internal market. By contrast, both fund managers could increase their utility if fund 2 were to sell some of its shares of asset C to fund 1.

[Insert Figure 1 about here]

Following the constrained max-min rule (10), manager $m1$ would seek to buy $|X| = 0.05$ shares of asset C from fund 2, as the resulting portfolio would maximize her indirect utility over all possible levels of cross-trading. Manager $m2$ would agree to this cross-trade because it would allow her to improve her utility relative to the NCT case. This case is identified by the vertical line OCT (which stands for optimal-cross-trading). In fact, fund 2 would be happy to sell even more shares of asset C to fund 1, because her utility increases even further to the right of OCT. However, $m1$ already attained her overall maximum at OCT, and will no longer participate in cross-trading beyond OCT. The resulting cross-trading is a mutually beneficial outcome for both fund managers. The existence of an internal market within the fund family improves both managers' utility.¹⁴

The purchase of 0.05 shares of asset C by fund 1 from fund 2 effectively deviates the initial portfolio composition of both funds from their benchmarks. In particular, it makes the liquid fund 1 more illiquid, and fund 2 more liquid. We next show that this pattern intensifies with the degree of benchmark style disparity, with important implications for the fund family and the delegating households.

¹⁴ Figure 1 also shows that this result is not unique to our assumption about funds' initial portfolio composition. We can obtain even higher levels of optimal cross-trading if we change the parameter values of our problem. If we keep the number of shares of asset C fixed at the family-level, any initial split of C between the two funds that moves NCT to the left in Figure 1 would result in a higher level of cross-trading ($|X| > 0.05$).

3.4 Style diversity, cross-trading, and portfolio liquidity

Figure 2 plots the optimal level of cross-trading (Panel A) and the associated average excess holdings of the illiquid asset (Panel B) as a function of style diversity.¹⁵

[Insert Figure 2 about here]

Cross-trading increases monotonically with style diversity. The optimal crossing goes from $|X| = 0.04$ for the lowest style diversity (corresponding to $\gamma_{h1} = 6$), to $|X| = 0.23$ for the largest style diversity we consider (which corresponds to $\gamma_{h1} = 9$). The intuition is that high style diversity leads managers to follow more extreme benchmark compositions and exposes them to insufficient diversification, with $m1$ having too little and $m2$ too much exposure to asset C . Managers then require larger rebalancing of their initial portfolios to achieve more diversified portfolios. Given that $m1$ demands more shares of C while $m2$ is willing to supply those shares, they can satisfy their need for rebalancing by trading with each other. The positive relation between cross-trading (offsetting trades) and style diversity within a family is one of the main predictions of our model. We find empirical support for this prediction in a sample of U.S. actively-managed mutual funds in subsection 5.2.

In contrast, insufficient diversity leads to little or no cross-trading. This would be the case, for instance, if the RRA coefficient of the households to whom liquid style L is catered was similar to the RRA coefficient of the manager (e.g., $\gamma_{h1} = 4$). Thus, intra-family style diversity not only strengthens the extent of cross-trading in our model, but is a necessary condition for managers to consider crossing their trades in the first place.

Panel B of Figure 2 shows that, as intra-family style diversity increases, each fund's deviation from the benchmark also increases. Moreover, each fund deviates in the direction of each other's benchmark. That is, *the FA fund that follows the more liquid style L (fund*

¹⁵ As argued in Section 3.1, style diversity is positively related to the RRA coefficient γ_{h1} of household $h1$. Therefore, we use γ_{h1} as our measure of diversity.

1) *increases its portfolio illiquidity more than an equivalent SA fund, while the converse is true for the fund that follows the illiquid style S (fund 2).*¹⁶ We empirically test this second main prediction of our model in subsection 5.3.

3.5 Implications for the overall fund family

Our model assigns no role to a centralized decision maker, such as a Chief Investment Officer (CIO) of the overall family.¹⁷ In practice, CIOs may not have a direct say on the investment decisions of the affiliated funds but still have the responsibility for the strategy of the overall group. Under the family arrangement, it could be argued that the CIO would preclude cross-trading (by, e.g., offering a narrow set of investment styles) unless it satisfies not only the interests of the fund managers but also the interests of the family. Thus, our results so far can only represent an equilibrium situation if the net benefits of cross-trading to the family are non-negative.

To assess these net benefits, we compare the expected growth of assets under management, measured as $R^*(T) \equiv \ln(W^*(T)/W(0))$, accruing to (the sum of) two FA funds versus (the sum of) two otherwise equivalent SA funds.¹⁸ Because the only difference between FA and SA funds is that the former are allowed to cross-trade, the difference between a family of two FA funds and a group of two SA funds represents the benefits to the family from

¹⁶ We would like to highlight that fund managers are less likely to trade in opposite directions over the investment period if they are both more (respectively, less) risk averse than the less (more) risk tolerant household $h1$ ($h2$). In this case, when underperforming their respective benchmarks both managers would want to deviate by decreasing the weight of their portfolios in the relatively riskier and illiquid asset C (see Basak, Pavlova, and Shapiro (2007)). In this sense, no-cross-trading (NCT) could also be an optimal outcome. Since this situation reflects cases of “extreme”—either low or high—risk aversions for the managers, we concentrate on the more plausible case of intermediate levels of risk aversion for both managers in the rest of our analysis.

¹⁷ In this sense, our setup differs from Binsbergen, Brandt, and Koijen (2008), who assume that a CIO allocates capital to the different investment styles. Although realistic in the context of, for instance, pension funds, their assumption does not necessarily hold in the context of open-end mutual funds for which the assets under management are decided by households. To reduce the layers of agency involved in our problem, we thus assume that no CIO makes investment decisions for the family-affiliated funds.

¹⁸ Since mutual funds derive revenue from the fees they collect on total assets under management, asset growth provides a direct measure of benefits to the fund management companies.

facilitating cross-trading. We present the results in Figure 3, which plots these benefits as a function of style diversity.

[Insert Figure 3 about here]

Across style diversity scenarios, the family derives positive benefits from allowing its affiliated funds to cross trades in the internal market. Moreover, the average growth in the AUM of the family increases monotonically with style diversity, from marginally positive values for low diversity (e.g., $\gamma_{h1} = 6$) to a few percentage points for large enough style dispersion (e.g., $\gamma_{h1} = 9$).

Since higher style diversity leads to more cross-trading in our model, the figure suggests that families can derive larger benefits from increasing the menu of investment styles available to investors. Thus, a centralized decision maker for a family of mutual funds (i) will not discourage internal trading across FA funds, and (ii) will offer a menu of investment styles that is as diverse as possible. The latter result seems to indicate that an optimal strategy for the family is to maximize the style diversity and encourage FA funds to follow as extreme benchmarks as possible. However, as we show in the next subsection, such a strategy can be severely limited by the costs it imposes on delegating households.

3.6 Delegation costs of cross-trading

To the extent that lower transaction costs translate into better net-of-fees performance, the availability of internal markets provides benefits to the investors in FA versus otherwise identical SA funds. However, internal markets can also be costly to FA fund investors. In particular, they can facilitate opportunistic trades, such as those induced by misalignment of incentives between managers and households, that would otherwise be too costly to execute in the open market as SA funds.

To determine how this tradeoff plays out in the model, we assess the net effects of portfolio delegation on the investors' utility by examining their certainty equivalent (CE)

rate of return. For an investment policy $\varphi = \{Z(0), Z(1)\}$, this is the risk-free rate of return $CE(\varphi)$ that makes an agent indifferent between following the policy φ over the investment period or alternatively earning this risk-free rate on the same initial investment over the same period. For an agent with CRRA coefficient γ and initial wealth w , $CE(\varphi)$ solves:

$$\frac{[CE(\varphi)w]^{1-\gamma}}{1-\gamma} = E \left[\frac{(W^\varphi(T))^{1-\gamma}}{1-\gamma} \right], \quad (11)$$

where the superscript denotes that the final wealth $W^\varphi(T)$ is attained under the investment policy φ .

We report the net effect of delegation to FA versus SA funds in Figure 4. The reported effects arise only due to the potential cross-trades under the family arrangement. Hence, they can equivalently be interpreted as *net effects of cross-trading*. We split the analysis in two.¹⁹ In Panel A, we report the net costs of delegation to FA funds attributable exclusively to the manager's risk-shifting behaviour in response to the convex incentives that the flow-performance relation (4) induces. In Panel B, we report the approximate benefits of cross-trading to delegating households. We compute these benefits by removing the convexity in the managers' incentives, as if the fund pursued a *passive* strategy, rather than an *active* one.²⁰ The manager of a passive fund has no incentive to engage in risk-shifting. Hence, the cross-trading between two FA passive funds should reflect only transaction cost-saving and diversification considerations. The difference between the effects in Panels A and B determines the net costs of delegation to the FA funds in our model.

[Insert Figure 4 about here]

The net effect of cross-trading on the households' utility can be substantially negative.

¹⁹ In all the cases, a household's derived utility under delegation is obtained by plugging her managed fund's terminal wealth in the utility function of equation (1).

²⁰ Specifically, to compute the CER for households who delegate to passive funds, we solve for the managers' optimal investment policies when $\phi(R_j^F(T) - R_j^Y(T)) = \phi^L + \psi \left[e^{R_j^F(T) - R_j^Y(T)} - e^\eta \right]$, for $j \in \{1, 2\}$. In other words, we remove the convex kink from expression (4).

For low style diversity ($\gamma_{h1} = 6$), consistent with relatively low levels of cross-trading (see subsection 3.4), Panel A shows that the net costs of risk-shifting to delegating households are zero, implying that FA funds do not engage in more risk-shifting behaviour than their SA counterparts. However, cross-trading allows FA fund managers to deviate from their benchmarks more than investors would like, resulting in negative (albeit small) net benefits of cross-trading in Panel B.

For the investors in either of the two funds, the costs of risk-shifting induced by cross-trading increase monotonically with style diversity. The benefits of cross-trading also increase monotonically for the investor in fund 2, and increase first but then decrease for the investor in fund 1. In both cases, however, the benefits increase to a lesser extent than the costs of cross-trading. As a result, high levels of style diversity can lead to substantially low CE returns to the delegating households of fund 2 and, especially, for the relatively more risk-averse household that invests in fund 1. Since cross-trading increases with diversity these results imply that, for large levels of cross-trading, managers' optimal policies can make *both households h1 and h2 worse-off after cross-trading*. The reason is that the managers' policies in response to their benchmarks expose the delegating households to inadequate—either excessive or insufficient—risks, with negative utility implications in both cases. These costs should limit, in industry equilibrium, the extent to which fund families could exploit the relation between style diversity and the benefits that we describe in subsection 3.5.

4 Discussion

Cross-Subsidization Hypothesis and Centralized vs. Decentralized Decisions.

Gaspar, Massa, and Matos (2006) conjecture, and empirically verify, that the convexity of funds' flow-performance relationship can encourage fund families to 'play favorites' among affiliated funds in order to maximize the family's total amount of assets under management.

In particular, they find that fund families can cross-subsidize some member funds over others through interfund transactions at *below or above market prices*. A novel prediction of our model is that families can still increase the amount of assets under management without playing favorites by choosing to offer diverse enough styles in their fund menus. The FA funds following these styles will indirectly satisfy the family objective while engaging in *mutually beneficial* cross trades, even without the intervention of a centralized decision maker.²¹

Our model can also be used to show how, in the presence of illiquid assets, families can play favorites among FA funds by having them cross trade *even at fair market prices*. We illustrate this point in Figure 5, which reports the level of cross-trading that a CIO of the overall family would choose if deciding to play favorites between funds 1 and 2 in our model.²² Starting from NCT, the CIO's derived utility increases in the number of shares of asset C that fund 2 sells to fund 1, up to OCIO. Hence, the CIO benefits by maximizing manager $m2$'s utility ($OM2$) rather than $m1$'s utility ($OM1$). At $OM2$, the return-to-risk ratio of fund 2 is higher than that of a comparable SA fund (NCT), while the return-to-risk ratio of fund 1 is significantly lower. Relative to the no-cross-trading (NCT) scenario, the resulting funds' measured performance resembles a "cross-subsidization" of fund 2 at the expense of fund 1. This type of cross-trading would still violate the spirit of Rule 17(a)-7, even if executed at *fair market prices*, because according to the rule both funds should benefit from the trade. In spite of the substantial negative effects accruing to delegating households, regulatory authorities would arguably find such breach of fiduciary duty hard to detect in practice.

²¹ Arguably, our model's assumption that fund managers seek to beat their benchmarks to increase compensation might not be a decentralized decision, but an outcome of the fund families' optimal incentive structure. This would be particularly the case if the payoff function (4) reflected the explicit incentive contract the family offers to the portfolio manager. This argument is less valid, however, when (4) reflects the implicit compensation function created by the households' flows as a function of funds' benchmark-adjusted performance.

²² For this figure, we assume that the CIO chooses the level of cross-trading that maximizes CRRA utility over end-of-period total assets under management $\sum_{j=1,2} \phi(R_j^F(T) - R_j^Y(T))F_j(T)$, for a RRA coefficient for the CIO equal to that of the fund managers, i.e. $\gamma_c = \gamma_m$.

[Insert Figure 5 about here]

Other Benefits and Costs of Family Affiliation. The results in subsection 3.6 suggest that delegating households could bear significant costs of cross-trading, and that these costs increase with the level of style diversity offered by the fund family. These results pose the question of why, in equilibrium, investors would delegate their money to FA funds instead of SA funds (or FA funds from families with low style diversity).

Even though this question cannot be addressed within the partial equilibrium framework of our paper, industry equilibrium considerations suggest that other benefits (and costs) of family affiliation will tend to make investors indifferent between FA and SA funds in practice. In particular, our model abstracts from potential benefits to fund shareholders of investing in a large and diverse family such as better allocation of managerial resources (Berk, van Binsbergen, and Liu (2017)) or lower fees (Khorana, Servaes, and Tufano (2009)). Moreover, our definition of illiquidity necessarily emphasize certain dimensions—e.g., transaction costs—more than others. Assume, for instance, that a manager had access to a private, short-lived signal, at the beginning of the investment period. These managers might find it optimal to trade aggressively on the signal early on, before its information content becomes public. Cross-trading could then facilitate not only pure savings on transaction costs but also fast order execution.²³ Even if it also aggravated risk-shifting problems (and associated costs), the benefits might outweigh the costs for valuable enough private signals.²⁴

Active vs. Passive Funds. Our analysis in this paper focuses on cross-trading between active mutual funds affiliated to a family. However, fund families often offer passive mutual funds as well, whose objective is to replicate the benchmark that defines their investment

²³ We thank an anonymous referee for suggesting this possibility.

²⁴ If managers had access to private signals, our definition of risk-shifting would need to be adjusted to reflect only excessive deviation—relative to what the investor with the same information would prefer—from the benchmark, beyond any deviation purely driven by information. Clearly, both types of deviations could intensify when cross-trading is possible in this potential extension of our model.

style. As we argue in subsection 3.6, passive funds do not respond to convex incentives, so the cost of cross-trading associated with risk-shifting behaviour that our model identifies should be approximately zero for this type of funds. However, FA passive funds could still trade with one another to save in transaction costs associated with tracking different style benchmarks. Hence, the possibility of cross-trading would mostly lead to benefits for passive funds. We would then expect diversity to have a positive impact on measured performance of passive funds. Panel B of Figure 6 shows that this is indeed the case, as (i) FA passive funds outperform, on average, otherwise equivalent SA funds, and (ii) the average outperformance increases with style diversity.

[Insert Figure 6 about here]

We empirically test the implications for fund performance illustrated in Figure 6. The results of those empirical tests are described in subsection 5.4.

We note that, in the absence of a centralized decision maker, our model will generally predict less cross-trading between an *active* and a *passive* fund, than between two active funds. The reason is that, for *decentralized* trading to take place, both managers need to agree to the cross-trade, and would do so to optimally deviate from their benchmarks. Since a passive manager's goal is to track its benchmark as closely as possible, unlike an active manager, she has fewer incentives to cross-trade and deviate.²⁵

5 Empirical Analysis

In this section, we test some of the novel implications of our model. First, we test the hypothesis that style diversity within a family is a first-order driver of interfund transactions,

²⁵ In the presence of a centralized decision maker our model can still predict a large extent of cross-trading whenever such trading offers benefits to the family. For instance, if cross-trading allowed the family's CIO to increase the overall assets under management, the CIO could facilitate the transaction even against the interests of the passive manager. In this case, a fraction of the cost of cross-trading would be borne by the passive fund investors and, under the assumptions above, by the passive fund manager as well.

controlling for other variables that could explain interfund dealing such as commonality in holdings and correlation of flows across funds within a family. Second, we test the hypothesis that style diversity leads fund managers following high- and low-liquidity styles to deviate from their benchmarks in predictable directions. Specifically, a high (low) liquidity fund is likely to decrease (increase) portfolio liquidity. Third, we provide some evidence of the trade-off between the benefits and costs of cross-trading by examining the fund performance implications of style diversity. To the best of our knowledge, the existing empirical evidence on mutual funds has not yet verified either one of these central implications of our model.

5.1 Sample statistics

We test the implications of our model using a sample of U.S. actively-managed mutual funds. Our final sample covers the period 1999-2009 for a total of 443 families and 1,855 funds. In Appendix A we provide a detailed description of the data sources, filters, and the construction of the variables used in our analysis. We summarize the definition of each empirical variable in Appendix B.

Table 2 provides descriptive statistics for the main variables used in our empirical tests. Panel A shows the statistics for the family-level variables. The median family in our sample has about 5 funds under its umbrella (NFUNDS) and \$5.7 billion in assets under management (FAMSIZE). The median ratio of the amount of offsetting trades to the total internal volume in a fund family is 1.6% when computing offsetting trades for all the stocks (FAMOFFSET) and 0.9% when computing offsetting trades for illiquid stocks only (FAMOFFSETILLIQ). The median value of common holdings within fund families (FAMCOMHOLD) in our sample is 0.354.

[Insert Table 2 about here]

Funds that follow different styles also hold a significant amount of stocks in common. It is perhaps not surprising that the commonality in holdings of liquid stocks across funds

(FAMCOMHOLDLIQ) is larger than that of illiquid stocks (FAMCOMHOLDILLIQ), but we note that there is also a sizeable degree of commonality in holdings of illiquid stocks across funds following different styles, which validates one of the central assumptions of our model.

Panel B reports summary statistics at the fund-level, whereas Panels D and E report the correlations among the family-level and the fund-level variables, respectively. Pearson correlations are reported above the diagonal, and the p-values are reported below the diagonal.

Panel C provides descriptive statistics for the placebo families that we analyse in subsection 5.2. To determine whether affiliation with a given family is what drives the relation between style diversity and offsetting trades, following Gaspar, Massa, and Matos (2006) we create a placebo family for each actual family in our sample by randomly pairing each member fund with a non-affiliated fund that follows the same benchmark. This guarantees that we keep the level of diversity within the placebo family similar to that in the actual family. Panel C shows that the difference in style diversity between actual and placebo families is statistically not significant (with a p-value of 0.238). However, actual and placebo families appear to differ in the other dimensions (e.g. offsetting trades for illiquid stocks only), which is expected because the placebo families are intended to break the connections between the member funds. However, these differences are not economically significant.²⁶

²⁶ Panel C of Table 2 shows that the number of funds and the size of the placebo families is smaller than those of the actual families. Even though these differences are statistically significant (with p-values of 0.008 and 0.009, respectively), they are not economically large. The difference of -0.056 in the number of funds represents only 1.2% of the average number of funds in the actual family (i.e. the mean NFUNDS of 4.709 in Panel A of Table 2). The difference of -\$353 million in family size represents 6.1% of the average family size in the actual family (i.e. the mean FAMSIZE of \$5,801 million in Panel A of Table 2). The reason for such differences is that, when creating placebo families by randomly drawing funds from other families, it is sometimes impossible to find a matching fund following the same benchmark.

5.2 Style diversity and offsetting trades in fund families

In order to test the hypothesis that style diversity within a family is a first-order driver of interfund transactions, we implement cross-sectional regressions using the following specification:

$$\text{FAMOFFSET} = a_0 + a_1 \text{FAMDIVERS} + a_2 \text{Controls} + \epsilon \quad (12)$$

where we control for other variables that could explain interfund dealing such as commonality in holdings (FAMCOMHOLD) and correlation of flows (FAMFLOWCORR) across funds belonging to the same family. We perform Fama and MacBeth (1973) regressions with heteroskedasticity-consistent standard errors and Newey and West (1987) correction using four lags. We present the results of this estimation in Table 3. In Panel A the dependent variable is the amount of offsetting trades across all the stocks traded within the family in a given quarter (FAMOFFSET), while in Panel B we replace the dependent variable in (12) with an offsetting measure that uses illiquid stocks only (i.e. FAMOFFSETILLIQ). The dependent variables are normalized using the total trading volume for a family-quarter-stock combination (see Appendix B for more details).

[Insert Table 3 about here]

As predicted, the coefficient for the main variable of interest, style diversity (FAMDIVERS), is positive and statistically significant at the 1% level in both Panels A and B. The economic magnitude of these results is also very strong. For instance, in column (1) of Table 3 (Panel A), a coefficient of 0.0544 for FAMDIVERS indicates that a one-standard-deviation increase in style diversity (which is 0.124 in Table 2, Panel A) is associated with an increase in offsetting trades (FAMOFFSET) by 0.0067 (i.e., the product of 0.0544 by 0.124). This value of 0.0067 needs to be compared with the average offsetting trades in Table 2 (Panel A) of 0.015. Therefore, a one-standard-deviation increase in intra-family style

diversity is associated with a 45% increase in the average amount of offsetting trades within the family.

In columns (2) to (4) of Table 3, we run univariate regressions of offsetting trades on other potentially related variables, such as commonality in holdings across the funds in the family (FAMCOMHOLD, column 2), the correlation of their money flows (FAMFLOWCORR, column 3), and the family-level incentive to cross-subsidize high-fee funds at the expense of low-fee funds (FAMFEEDISP, column 4). The coefficients on these variables are all significant at the 1% level and have the expected sign: (1) the larger the extent of common holdings across the sibling funds, the more likely they are to eventually trade the same stocks in opposite directions (i.e., a positive sign on FAMCOMHOLD), (2) the lower the correlation of flows across sibling funds, the more they will be induced to trade in opposite directions (i.e., a negative sign on FAMFLOWCORR), and (3) the larger the dispersion in fees charged by the funds within a family, the more likely it is that the family will want to transfer performance from low-fee funds to high-fee funds through cross-trading, as hypothesized by Gaspar, Massa, and Matos (2006) (i.e., a positive sign on FAMFEEDISP).

In columns (5) to (7) and (9) of Table 3, we show that the effect of style diversity on a family's offsetting trades is not explained away by commonality of holdings (FAMCOMHOLD), fund flow correlations (FAMFLOWCORR), or dispersion in fees (FAMFEEDISP), either taken individually or controlling simultaneously for all of them. The coefficient on FAMDIVERS remains statistically significant at 1% level, and economically large. In column (9) of Table 3 (Panel A), i.e. the full model with all the controls, a one-standard-deviation increase in style diversity is associated with a 21% increase in the family's offsetting trades. In comparison, the economic significance of the remaining controls is substantially smaller: a one-standard-deviation increase, respectively, in commonality of holdings, fund flow correlations and dispersion in fees, is associated with a change in the family's offsetting trades of 7.3%, -0.8%, and 0.75% of the average offsetting trades across families. These results suggest

that our measure of intra-family style diversity, FAMDIVERS, is of first-order importance in explaining the degree of offsetting trades within the family.

When we restrict our analysis to the offsetting trades of illiquid stocks only (i.e., Panel B of Table 3), and after including all the control variables (i.e., column (9)), a one-standard-deviation increase in style diversity is associated with a 25% increase in the family's offsetting trades in the relatively more illiquid stocks (FAMOFFSETILLIQ).²⁷

In column (10) of Panels A and B of Table 3 we run the multivariate regression of column (9) on the placebo families. In contrast to the actual families, intra-family style diversity has no statistically significant relation with the level of offsetting trades for either all stocks (Panel A) or the relatively more illiquid stocks (Panel B).

Overall, the results in Table 3 confirm our model's main prediction that the diversity in styles offered by a mutual fund family is a first-order determinant of, and is positively related to, the offsetting trades within the family.²⁸

5.3 Style diversity and distortions in funds' portfolio liquidity

Another main implication of our model is that a fund following a high-illiquidity benchmark is likely to deviate from the benchmark by decreasing portfolio illiquidity, and conversely for a fund following a low-illiquidity benchmark. Given the potentially negative consequences of this behaviour on delegating households' utility, as we emphasized in subsection 3.6, we

²⁷ This is obtained by multiplying 0.124 from Table 2 with the coefficient of 0.0182 for FAMDIVERS from Table 3, Panel B, column (9), and dividing by the average FAMOFFSETILLIQ of 0.009 from Table 2, Panel A.

²⁸ We have also performed the analysis of Table 3 using panel regressions with year-quarter fixed effects and standard errors clustered at the family and year-quarter levels, and the results are qualitatively unchanged. For brevity, we have not tabulated the panel regressions, but the results are available upon request.

test the validity of this prediction using the following specification:

$$\begin{aligned} \text{FUNDILLIQDEV} = & \alpha_0 + \alpha_1 \text{DMEASURE} + \alpha_2 \text{LOWBILLIQ} + \alpha_3 \text{HIGHBILLIQ} + \\ & + \alpha_4 \text{DMEASURE} \times \text{LOWBILLIQ} + \alpha_5 \text{DMEASURE} \times \text{HIGHBILLIQ} + \\ & + \alpha_6 \text{Controls} + \psi, \end{aligned}$$

where the dependent variable FUNDILLIQDEV is the difference between the fund illiquidity and its benchmark illiquidity. This variable can take positive or negative values. DMEASURE denotes a family diversity measure, which we capture using alternately the number of affiliated funds (NFUNDS), the amount of assets under management (FAMSIZE), or the minimum of minus one times the correlation of intra-family benchmark returns (FAMDIVERS). The variable LOWBILLIQ (HIGHBILLIQ) is an indicator function that equals one if the fund benchmark ranks in the bottom (top) tercile of illiquidity across all the benchmarks in a given quarter. Controls is a vector of other determinants of the deviation in liquidity between the fund and its benchmark (FUNDILLIQDEV). In particular, we control for the age (in years) of the fund since inception (FUNDAGE) and the TNA of the fund (FUNDSIZE). These two variables are very skewed. We use a logarithmic transformation of those variables to deal with skewness. We also include indicator variables DG and DGI, which equal one if the fund is in the 'Growth' or the 'Growth and Income' investment categories, respectively, and equals zero otherwise. Our controls also include the past 12-month volatility of the fund's monthly returns (FUNDRETVOL), the fund's expense ratio (FUNDEXPRATIO), a dummy that equals one if the fund ranked in the bottom half across all the funds in the same investment category in the previous quarter (DLOSER), and the average illiquidity of the benchmarks within the fund's family (FAMBENCHILLIQ).

Our model predicts a positive sign for the interaction between DMEASURE and LOWBILLIQ, which means that funds following low-illiquidity benchmarks deviate by increasing

their portfolio illiquidity as the family diversity measure increases. Similarly, the model prediction for the interaction between DMEASURE and HIGHBILLIQ is negative, as funds following high-illiquidity benchmarks are expected to deviate by decreasing their portfolio illiquidity as the family diversity measure increases. We are especially interested in the interactions involving our style diversity measure FAMDIVERS. We then run Fama and MacBeth (1973) regressions with heteroskedasticity-consistent standard errors and Newey and West (1987) correction using four lags. We report the results of the estimation of this model in Panel A of Table 4.

[Insert Table 4 about here]

As predicted, funds following low-illiquidity benchmarks increase their portfolio illiquidity more as their family style diversity increases (the coefficient on the interaction between FAMDIVERS and LOWBILLIQ is positive), whereas funds following high-illiquidity benchmarks decrease their portfolio illiquidity more as their family style diversity increases (the coefficient on the interaction between FAMDIVERS and HIGHBILLIQ is negative), with both effects being statistically significant at the 1% level. The results using the two alternative measures of intra-family diversity (i.e. NFUNDS and FAMSIZE) are not as significant as the results using FAMDIVERS. In column (4), we include all the three diversity measures in the same specification, as well as their multiple interactions with LOWBILLIQ and HIGHBILLIQ. In the full specification, only the interaction terms involving FAMDIVERS remain robustly significant at the 1% level. These findings confirm the importance of style diversity in the choice of portfolio liquidity for family-affiliated funds, in agreement with our theory model.

To further assess the relevance of our model to explain these empirical findings, we next test whether our results are driven, following the cross-subsidization hypothesis, by *family-coordinated*—as opposed to *decentralized*—trades between affiliated funds. Specifically, we test the hypothesis that the deviations in funds' illiquidity from their benchmarks' illiquidity

is a consequence of the cross-subsidization of high-value (high-fee) funds at the expense of their low-value (low-fee) siblings within the family.

In Panel B of Table 4, we re-run the specification of column (3) in Panel A with two additional indicator variables, LOWFEE and HIGHFEE, their interactions with LOWBILLIQ and HIGHBILLIQ, as well as triple interactions involving FAMDIVERS. The variable LOWFEE (HIGHFEE) is an indicator that equals one if the fee charged by the fund is in the bottom (top) tercile of the fees charged across all the funds within a given family.

Our estimates indicate that we can rule out this alternative explanation. If the effect we document in column (3) of Panel A in Table 4 was the result of a strategy played at the family-level, we would expect to observe (i) that deviations in funds' liquidity from their benchmarks are stronger for high- and low-value funds, both before and after controlling for the family's style diversity (FAMDIVERS), and (ii) that style diversity cannot explain deviations in funds' liquidity after controlling for high- and low-value fund effects. By contrast, we find that whether a fund has low or high value to the family does not help explain its liquidity relative to the benchmark—the coefficients on LOWFEE and HIGHFEE, and their interactions with LOWBILLIQ, HIGHBILLIQ and FAMDIVERS are all statistically not different from zero in columns (1) and (2). However, style diversity remains a strong determinant after controlling for a fund's value to the family—the coefficients on the interactions $\text{FAMDIVERS} \times \text{LOWBILLIQ}$ and $\text{FAMDIVERS} \times \text{HIGHBILLIQ}$ retain their predicted signs and statistical significance. We conclude that our empirical findings cannot be attributed to the cross-subsidization hypothesis, lending further support for the main mechanism in our model.

5.4 Implications for fund performance

Our model suggests that cross-trading has both benefits and costs. It allows funds to save on transaction costs, but it also encourages risk-shifting, which can be costly to fund investors.

In this section, we look into the benefits and costs of cross-trading by examining its implications for fund performance. We implement cross-sectional regressions using the following specification:

$$\text{FRETADJ}(q+1) = a_0 + a_1 \text{FAMDIVERS}(q) + a_2 \text{Controls}(q) + \epsilon \quad (13)$$

where $\text{FRETADJ}(q+1)$ denotes the benchmark-adjusted performance of the fund in quarter $q+1$. The style diversity (FAMDIVERS) and control variables are measured at quarter q .

We use the following strategy to distinguish the benefits of cross-trading from its costs. Following our argument in Section 4, we assume that funds that commit to tracking their benchmarks very closely (e.g., passive funds) only engage in cross-trading for the benefits it provides, not for risk-shifting purposes. For these funds, our model predicts that the average benchmark-adjusted performance should increase with style diversity, as illustrated in Panel B of Figure 6. By running regression (13) on a subsample of *passive* funds, we then expect to capture only the benefits of cross-trading.

By contrast, according to our theory model, active funds (i.e., those that deviate significantly from their benchmarks) use cross-trading not only for saving on trading costs but also for risk-shifting purposes. Therefore, the expected impact of cross-trading on the performance of active funds is, *a priori*, less clear. In fact, Panel A of Figure 6 indicates that funds following relatively liquid styles experience net benefits, while funds following relative illiquid styles experience net costs, and that both benefits and costs of cross-trading can increase with style diversity. However, the model-implied average impact of cross-trading on the benchmark-adjusted performance across the active funds in the family is approximately zero at all levels of diversity. This outcome is consistent with our industry equilibrium argument of Section 4, whereby fund families should engage in cross-trading up to the point in which the benefits to their investors offset the associated risk-shifting costs. Accordingly, when

running regression (13) on a subsample of active funds we expect to find an insignificant net effect of FAMDIVERS(q) on FRETADJ(q+1).

We implement our performance test by ranking all funds in each cross-section into terciles according to the active share measure of Cremers and Petajisto (2009). Next, we create two subsamples, one that includes only the funds in the bottom tercile of active share in each cross-section (the “passive” funds), and one that includes only the funds in the top tercile of active share in each cross-section (the “active” funds). Table 5 shows the regression results separately for these two subsamples.

[Insert Table 5 about here]

Columns (1)-(2) report the results for the subsample of passive funds, and columns (3)-(4) for the subsample of active funds. In both cases, the results are broadly consistent with our predictions. Indeed, column (1) shows that an increase in a family’s style diversity (FAMDIVERS) leads to an improvement in the average performance of passive funds. The coefficient on FAMDIVERS is positive and significant at the 10% level. By contrast, column (3) indicates that the effect of style diversity on the average performance of active funds is statistically indistinguishable from zero.

In columns (2) and (4) we separate passive and active funds by the liquidity of their benchmarks, using the following regression specification:

$$\begin{aligned} \text{FRETADJ}(q+1) = & \alpha_0 + \alpha_1 \text{FAMDIVERS}(q) + \alpha_2 \text{LOWBILLIQ} + \alpha_3 \text{HIGHBILLIQ} + \\ & + \alpha_4 \text{FAMDIVERS} \times \text{LOWBILLIQ} + \alpha_5 \text{FAMDIVERS} \times \text{HIGHBILLIQ} + \\ & + \alpha_6 \text{Controls} + \psi, \end{aligned}$$

The results of this test provide additional support for the model-implied pattern of benchmark-adjusted performance in Figure 6. Unconditionally, style diversity remains strongly

positively associated with the performance of passive funds but has no impact on the performance of active funds. Conditional on the illiquidity of a fund benchmark (i.e. $\alpha_1 + \alpha_4$ or $\alpha_1 + \alpha_5$), the performance of passive funds improves with family style diversity irrespective of benchmark illiquidity (i.e. column (2)). By contrast, the performance of active funds following low-illiquidity benchmarks improves while the performance of active funds following high-illiquidity benchmarks deteriorates as style diversity increases (i.e. column (4)).

6 Conclusion

We propose a model of internal markets in fund families and explore the possibility that benchmark concerns generate opportunistic internal trades between family-affiliated funds. When the family offers a diverse enough menu of investment styles, and these styles represent portfolios with common holdings of illiquid assets, cross-trading can be the optimal outcome of a decentralized strategy implemented at the individual fund-level. Fund managers have an incentive to distort their portfolios away from their style benchmarks in an attempt to attract investors' flows. If the styles that the funds follow are sufficiently different from one another, managers will likely trade common holdings in opposite directions. Funds that belong to the same family can then use their access to the internal market to cross those opposite trades at low cost, instead of having to deal with illiquid open markets. This possibility allows them to aggressively change the liquidity of their portfolios with respect to their benchmarks, effectively engaging in more risk-shifting than their standalone counterparts. While the overall family stands to benefit from this strategy, the increased risk-shifting behaviour creates additional delegation costs to fund investors.

We provide empirical evidence consistent with the novel implications of our model. In particular, using a sample of U.S. actively managed equity mutual funds, we verify the hypothesis that style diversity within a fund family is positively associated with (i) the

amount of intra-family offsetting trades, and (ii) the degree to which funds deviate the liquidity of their portfolios from the liquidity of their style benchmarks.

Our results have several implications. First, they draw attention to a novel delegation cost associated to family affiliation and to a subtle, although economically significant, potential breach of fiduciary duty of family-affiliated fund managers with respect to their shareholders. Second, they provide an alternative rationale for the diversity in investment styles within fund families that we observe in practice. Finally, our results point to style diversity within fund families as a potential source of cross-sectional variation in the liquidity management by mutual funds.

Our model also has some limitations. In order to isolate the effects of decentralized intra-family trading on the choices of the individual funds, we abstract from decisions made at the family level. These include the choice of the number of funds and the menu of styles to offer by the family, which we take as exogenous in our model. This style diversity can lead to delegation costs to fund investors. The observed differences in style diversity across fund families could be explained as an equilibrium outcome. A model that would take into account differences in managerial skill, or differences in the determination of fund fees, could help explain the observed differences in style diversity across fund families. We leave this extension for future research.

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Appendix

A Data Sources, Filters, and Variables

We obtain fund returns, investment objectives, fees, total net assets (TNA), and other fund characteristics from the Center for Research in Security Prices (CRSP) Survivor-Bias-Free Mutual Fund Database. We use the Wharton Research Data Services MFLINKS file to merge this database with the Thomson Financial Mutual Fund Holdings dataset, which contains information on stock positions of funds and identifies fund families (Wermers (2000)). We restrict our analysis to diversified domestic actively managed equity mutual funds.²⁹ We compute fund-level variables by aggregating across all the share classes. For instance, a fund's expense and turnover ratios are the TNA-weighted averages of the ratios of its different share classes. The age of the fund is equal to the age of its oldest share class, and the TNA of the fund is the sum of the TNAs of all its share classes.

Following Kacperczyk, Sialm, and Zheng (2008) and Glode (2011), we exclude funds that hold on average less than 80% or more than 105% of their net assets in equity. This is because reported fund objectives do not always accurately characterize a fund. We also exclude funds with TNA of less than \$15 million as Elton, Gruber, and Blake (2001) show that the returns on such small funds tend to be biased upwardly in the CRSP database. In addition, we remove from our final sample observations associated with funds that at a given point in time have less than 18 months since inception, to minimize the effects of incubation bias (Evans (2010)). We also exclude funds that hold less than 10 stocks in their portfolios in a given quarter. We exclude fund families with less than two funds, as we need at least two funds to compute some of the pairwise measures used in our tests, which we describe in more detail below. Our final sample covers the period 1999-2009 for a total of 443 families and 1,855 funds. The time period of our sample is limited below by the CRSP daily mutual fund returns data, which starts in 1999, and is limited above by the active share sample of Cremers and Petajisto (2009) and Petajisto (2013), which ends in 2009.³⁰ Due to the quarterly frequency of mutual fund holdings data, we run our tests at the quarterly level.

²⁹ More specifically, and following existing literature, we include in our sample funds with AGG, GMC, GRI, GRO, ING, and SCG Strategic Insight codes, EIEI, G, LCCE, LCGE, LCVE, MCCE, MCGE, MCVE, MLCE, MLGE, MLVE, SCCE, SCGE, and SCVE Lipper codes, and G, G-I, AGG, GCI, GRI, GRO, LTG, MCG, and SCG Wiesenberger codes. We exclude international, balanced, sector, bond, money market, and index funds.

³⁰ Even if our model predictions are borne in the data, we expect our empirical findings to weaken after the introduction of SEC rules 38a-1 and 206(4)-7 (introduction of chief compliance officer independent from fund management) and the amendments to rule 204-2 (advisers must maintain copies of their compliance policies and procedures and copies of any records documenting the adviser's annual review of those policies) in 2004, see Eisele, Nefedova, Parise, and Peijnenburg (2017).

We construct a number of variables to help us perform our tests, some of which are computed at the family-level and others at the fund-level. We provide a detailed description of these variables in Appendix B.

First we describe our family-level variables. Two natural proxies for the style diversity offered by a fund family are the number of funds, which we denote as NFUNDS, and the total value of assets under management (FAMSIZE) of the family. However, it is possible that a small two-fund family offers more diverse (e.g., one very conservative and one very risky) investment styles compared to a large and relatively homogeneous—in terms of offered investment styles—multi-fund family. If this is the case, NFUNDS or FAMSIZE could fail to properly capture the extent of internal dispersion in fund styles. We construct an alternative measure, FAMDIVERS, to better reflect the style diversity offered by a fund family in our model. Specifically, for each family we compute all the pairwise correlations of returns across all the different benchmarks followed by the funds affiliated with that family. We take the minimum of all those pairwise correlations and denote that measure as FAMDIVERS.³¹ Because a benchmark portfolio is outside of the family's control, we avoid in this way the endogeneity issues that would arise if pairwise correlations were computed on fund returns instead.

According to our model, style diversity within a fund family (i.e. FAMDIVERS) is an important driver of interfund transactions. In order to test this prediction we create a proxy for interfund trading, FAMOFFSET, that captures the dollar amount of all buy and sell trades of a particular stock that gets offset within a family in a given quarter, as a fraction of the intra-family trading volume. Similarly, we create a measure of offsetting trades of illiquid stocks only, FAMOFFSETILLIQ. We follow the standard in the literature and use changes in portfolio holdings between adjacent quarters to proxy for the trading of mutual funds. We control for other variables that could explain the extent to which trades get offset within fund families. In particular, one should expect offsetting trades to be more likely in families with more common holdings and lower flow correlations across their affiliated funds. We capture these two dimensions by creating the measures FAMCOMHOLD and FAMFLOWCORR, respectively. One possible concern with FAMCOMHOLD is that it might be mechanically related to the degree to which funds within the family follow the same style. To examine whether funds with different styles also hold many stocks in common, we compute two additional measures of cross-fund overlap in stock holdings: FAMCOMHOLDLIQ and FAMCOMHOLDILLIQ. In the calculation of both measures, we restrict our pairs of

³¹ We take the negative of the minimum across all the pairwise correlations of benchmark returns within the family so that a decrease in that minimum correlation can be interpreted as an increase in style diversity within the family.

funds to follow distinct DGTW benchmarks. Moreover, FAMCOMHOLDLIQ includes only liquid portfolio holdings (i.e. stocks in the bottom quintile of the liquidity cost estimates of Hasbrouck (2009), in the cross-section of stocks held by mutual funds), while the calculation of FAMCOMHOLDILLIQ uses only illiquid portfolio holdings (i.e. stocks not in the bottom quintile of liquidity cost). The stocks we classify as liquid represent an average 43% of funds' equity portfolios, and their average liquidity cost is about one fourth the liquidity cost of the stocks we classify as illiquid.³²

Our testable predictions result from decentralized strategies at the level of the individual fund managers. However, Gaspar, Massa, and Matos (2006) document that families can also play strategies at a centralized level. In particular, they show that families can play favourites and maximize the amount of assets for the whole group by transferring performance from low-value funds to high-value funds ('cross-fund subsidization') via cross-trading. Gaspar et al. (2006) consider high-value funds to be young funds, small funds, and funds that charge higher fees. Because age and size are endogenous determinants of the flow-performance relationship according to Berk and Green (2004) and Huang, Wei, and Yan (2007), and this relationship is a key ingredient of our model, we distinguish high- and low-value funds based on their total fees only. More precisely, we assess the extent to which offsetting trades within fund families are driven by cross-subsidization by controlling for the dispersion of fees FAMFEEDISP across all the funds in a family-quarter combination.

Next, we describe the fund-level variables that we use to test the prediction that high-liquidity funds decrease their portfolio liquidity and vice-versa for low-liquidity funds. We create the measure FUNDILLIQDEV to capture the difference between the illiquidity of the fund's portfolio and the illiquidity of its benchmark.³³ The illiquidity of the fund's portfolio and the illiquidity of the benchmark are value-weighted averages of the illiquidity of their respective holdings.

We assign a benchmark to each fund-quarter according to the following procedure, which we explain in more detail in Appendix C. First, we obtain the fund's prospectus benchmark from Antti Petajisto's website.³⁴ To approximate the portfolio stock holdings of each prospectus benchmark, we assign to a fund-quarter the Daniel, Grinblatt, Titman, and Wer-

³² More precisely, the average Hasbrouck's effective liquidity costs are 0.19% and 0.69% for the liquid and illiquid stocks in our sample.

³³ Following Lou (2012), we use the liquidity cost estimates of Hasbrouck (2009) as our measure of illiquidity.

³⁴ <http://www.petajisto.net/data.html>. The list of official benchmarks that mutual funds disclose in their prospectuses in Antti Petajisto's website includes 5 indices from S&P, 12 indices from Russell, and 2 indices from Dow Jones / Wilshire. All the variables in this dataset are defined and the sample of funds is selected as in Petajisto (2013). The sample used in Petajisto (2013) covers the period 1980-2009.

mers (1997) (DGTW) benchmark that has a current quarter tracking error closest to the tracking error that Petajisto (2013) obtained when using the return series of the funds and their respective prospectus benchmarks. We believe that, even if the matching of DGTW benchmarks is made with error, there is no reason for a bias to exist in favour of our results.³⁵

We identify control variables that may jointly affect FAMDIVERS and FUNDILLIQDEV when testing the hypothesis that style diversity is an important driver of the difference between fund and benchmark illiquidity. Specifically, we control for a number of fund characteristics, including the value of assets under management (FUNDSIZE), the expense ratio (FUNDEXPRATIO), the number of years since inception (FUNDAGE), the volatility of returns (FUNDRETVOL), and the average illiquidity of the benchmarks within the family (FAMBENCHILLIQ).

³⁵ We adopt this indirect method to identify the funds' benchmarks because we do not have access to the index constituents of all the prospectus benchmarks reported in Petajisto (2013). We believe that this allows us to obtain a reasonable match of stock characteristics between the fund and its benchmark.

B Variable Definitions

Variable	Definition
<i>DG</i>	An indicator variable that equals one if the investment style of the fund is “Growth”, and equals zero otherwise.
<i>DGI</i>	An indicator variable that equals one if the investment style of the fund is “Growth and Income”, and equals zero otherwise.
<i>DLOSER</i>	An indicator variable that equals one if the performance of the funds ranks below the median across all the funds in the same style in a given quarter.
<i>FAMBENCHILLIQ</i>	The cross-sectional average illiquidity of the benchmarks followed by all the funds affiliated with the same family in a given quarter. The illiquidity of each benchmark is a value-weighted average of the liquidity cost estimates of Hasbrouck (2009) across all the stocks in the benchmark’s portfolio.
<i>FAMCOMHOLD</i>	The measure of common holdings in Elton, Gruber, and Green (2007) but only for the funds affiliated with the same family. For any pair of funds (A, B), it is the sum of the minimum fraction of the portfolio held in any stock i between the two funds, i.e., $COM(A, B) = \sum_i \min(X_{Ai}, X_{Bi})$, where X_{Ai} (X_{Bi}) is the value of the portfolio of fund A (B) that is invested in stock i , as a fraction of the total identifiable amount of common stock held in the portfolio of fund A (B).
<i>FAMCOMHOLDILLIQ</i>	Similar to the measure FAMCOMHOLD above, but only for pairs of funds with different style benchmarks, and for illiquid stock holdings only (i.e., stocks ranked in quintiles 2 to 5 of the liquidity cost estimates of Hasbrouck (2009) in the cross-section of stocks held by mutual funds).
<i>FAMCOMHOLDLIQ</i>	Similar to the measure FAMCOMHOLDILLIQ described above, but only for pairs of funds with different style benchmarks, and for liquid stock holdings only (i.e., ranked in quintile 1 of the liquidity cost estimates of Hasbrouck (2009) in the cross-section of stocks held by mutual funds).
<i>FAMDIVERS</i>	The minimum pairwise correlation of DGTW benchmark returns across all the distinct pairs of benchmarks followed by the funds affiliated with the same family, using rolling windows of 6 months of daily DGTW benchmark returns.
<i>FAMFEEDISP</i>	The cross-sectional standard deviation of expense ratios across all the funds affiliated with the same family in a given quarter.
<i>FAMFLOWCORR</i>	The average pairwise correlation of monthly percentage flows across all funds in the same family, using rolling windows of 36 months and requiring a minimum of 20 observations. Monthly flows are computed as follows: $FLOW(t) = [TNA(t) - (1 + RET(t)) \times TNA(t-1)] / TNA(t-1)$, where $TNA(t)$ is the value of assets under management at month t , $TNA(t-1)$ is the value of assets at the end of month $t-1$, and $RET(t)$ is the return of the fund during month t .

<i>FAMOFFSET</i>	The dollar amount of offsetting buy and sell trades of a given stock (i.e., the minimum between positive holdings changes for that stock aggregated across all funds in the family, and the absolute value of negative holdings changes for that same stock aggregated across all funds in the family – changes in holdings being computed using adjacent quarters), normalized by the total dollar trading volume on that stock within the family (i.e., the sum of all positive and (the absolute value of) negative holdings changes for that stock across all funds in the family), averaged across all the stocks traded by the funds in the family for each quarter.
<i>FAMOFFSETILLIQ</i>	Similar to the measure FAMOFFSET but for illiquid stocks only (i.e., stocks ranked in quintiles 2 to 5 of the liquidity cost estimates of Hasbrouck (2009) in the cross-section of stocks held by mutual funds in a given quarter).
<i>FAMSIZE</i>	The sum of the quarter-end monthly TNAs of all the funds affiliated with the same family.
<i>FRETADJ</i>	The return of a mutual fund minus the return of its DGTW benchmark in a given quarter.
<i>FUNDILLIQDEV</i>	The difference between the illiquidity of the fund portfolio and the illiquidity of its benchmark portfolio (i.e., the value-weighted average of the liquidity cost estimates of Hasbrouck (2009) across all the stocks in the portfolio in a given quarter).
<i>FUNDAGE</i>	The number of years since the inception of the fund until the current quarter. This is equivalent to the age of the fund's oldest share class, for multi-share class funds.
<i>FUNDEXPRATIO</i>	The expense ratio of the fund, which is computed as the TNA-weighted average of the expense ratios across all its share classes, in the case of multi-share class funds.
<i>FUNDRETVOL</i>	The standard deviation of fund returns for the prior 12 months ending in the quarter prior to the quarter under consideration (i.e., from month $t - 14$ to month $t - 3$ relative to the last month t of the quarter under consideration).
<i>FUNDSIZE</i>	The TNA of the fund in a given quarter, which is the sum of the TNAs across all the fund's share classes in the case of multi-share class funds.
<i>HIGHBILLIQ</i>	An indicator variable that equals 1 if the value-weighted liquidity cost of the benchmark portfolio ranks in the top tercile across all the benchmarks in a given quarter, and equals zero otherwise.
<i>HIGHFEE</i>	An indicator variable that equals 1 if the expense ratio of the fund ranks in the top tercile across all the funds belonging to the same family in a given quarter, and equals zero otherwise.
<i>LOGFAMSIZE</i>	The logarithmic transformation of the variable FAMSIZE.
<i>LOGNFUNDS</i>	The logarithmic transformation of the variable NFUNDS.
<i>LOWBILLIQ</i>	An indicator variable that equals 1 if the value-weighted liquidity cost of the benchmark portfolio ranks in the bottom tercile across all the benchmarks in a given quarter, and equals zero otherwise.
<i>LOWFEE</i>	An indicator variable that equals 1 if the expense ratio of the fund ranks in the bottom tercile across all the funds belonging to the same family in a given quarter, and equals zero otherwise.
<i>NFUNDS</i>	The number of funds affiliated with the same fund family in a given quarter.

C Benchmark Assignment Methodology

We assign a benchmark to each fund-quarter according to the following procedure. First, we create the daily series of returns for each of the 125 DGTW benchmarks, following the methodology in Daniel, Grinblatt, Titman, and Wermers (1997). Some of the key steps of the methodology in Daniel, Grinblatt, Titman, and Wermers (1997) involve a triple sort of common stocks into buckets of size, book-to-market (B/M), and momentum (quintiles for each of these three dimensions), in the month of July of each year. Moreover, it requires availability of COMPUSTAT data for at least 2 years, a minimum of 6 months of returns in CRSP, size weights are constructed using the market value in June (using the NYSE size breakpoints), the B/M ratio uses the market cap in prior fiscal-year end (and are adjusted with industry averages, using the 49 Fama-French industry classification available in Ken French's webpage), and the momentum factor is the 12 month return with one month reversal gap. More details on the methodology are available in Russ Wermers' website: <http://www.smith.umd.edu/faculty/rwermers/ftpsite/Dgtw/coverpage.htm>.

Second, we compute the pairwise tracking errors (annualized) between the daily returns of the largest share class of each fund (following Petajisto (2013)) and all the 125 DGTW portfolios. Finally, we assign to a fund the DGTW benchmark that has a tracking error closest to the tracking error that Petajisto (2013) obtained when using the return series of the funds and their respective prospectus benchmarks. The list of official benchmarks that mutual funds disclose in their prospectuses is available in Antti Petajisto's website for the period between 1980 and 2009: <http://www.petajisto.net/data.html>. It includes 5 indices from S&P, 12 indices from Russell, and 2 indices from Dow Jones / Wilshire. All the variables in this dataset are defined and the sample of funds is selected as in Petajisto (2013). The sample used in Petajisto (2013) covers the period 1980-2009. Following Petajisto (2013), we use quarterly rolling windows of 6-month daily returns in the computation of such tracking errors.

D Tables and Figures

Table 1: Mutual Fund Structure in the Model

This table summarizes the main ingredients of our theory model, which consists of two mutual funds that follow two different investment styles. These investment styles are associated with different performance benchmarks and different sets of assets. Each fund holds two risky assets, one perfectly liquid, and the other illiquid. They both hold the illiquid asset. Each fund is managed by a different manager and caters its product to a different investor/household.

Mutual Fund	Investment Style	Benchmark	Invests in Assets	Specialized in Asset	Manager	Catered to Household
F_1	L	Y_1	1, C	1	$m1$	$h1$
F_2	S	Y_2	2, C	C	$m2$	$h2$

Table 2: Summary Statistics

This table provides summary statistics for the main variables used in this study. Panels A, C and D report the summary statistics for the variables calculated at the family-level (actual or placebo). Those family-level variables include (1) NFUNDS, which is the number of funds affiliated with the same family, (2) FAMSIZE is the sum of the assets under management across all the funds affiliated with the same family, (3) FAMOFFSET is the minimum between aggregate buy and sell trades of a given stock across the funds in a family, divided by the total trading volume of that stock in that family (i.e., the sum of the aggregate buy and sell trades for that stock), averaged across all the stocks traded in the family in a given quarter, (4) FAMOFFSETILLIQ is similar to FAMOFFSET, but restricting the calculations to illiquid stocks only (i.e., stocks not in the bottom quintile of liquidity cost across all the stocks held by mutual funds in a given quarter), (5) FAMCOMHOLD captures the extent of commonality in holdings across the funds in the same family, following Elton, Gruber, and Green (2007), (6) FAMFLOWCORR is the average pairwise correlation of monthly flows across all the funds in the family, using rolling windows of 36 months, (7) FAMDIVERS is the intra-family style diversity measure (-1 times the minimum of the pairwise correlations of returns across all the benchmarks in a given family), (8) FAMFEEDISP is the cross-sectional standard deviation of expense ratios across the funds in the family in a given quarter, (9) FAMBENCHILLIQ is the average illiquidity across the benchmarks within a family, (10) FAMCOMHOLDLIQ is similar to FAMCOMHOLD but applied only to funds with distinct benchmarks and to liquid stocks (i.e., stocks in the bottom quintile of liquidity cost across all the stocks held by mutual funds in a given quarter), and (11) FAMCOMHOLDILLIQ is similar to FAMCOMHOLDLIQ but applied to illiquid stocks (i.e., stocks not in the bottom quintile of liquidity cost). Panels B and E, report the summary statistics for the variables calculated at the fund-level, which include (1) FUNDILLIQDEV, which is the difference between the illiquidity of the fund and the illiquidity of its benchmark, (2) FUNDEXPRATIO is the fund's expense ratio, (3) FUNDAGE is the number of years since the inception of the fund, (4) FUNDSIZE is the fund's TNA, and (5) FUNDRETVOL is the monthly volatility of fund returns, for the prior 12 months. The 'placebo' families of Panel C are constructed as follows: for each family in our sample, we randomly pair each member fund with a fund that follows the same benchmark but does not belong in the same family (without replacement). Panels D and E report the correlations between family-level and fund-level variables, respectively. Pearson correlations are reported above the diagonal, and their respective p-values are reported below the diagonal.

PANEL A: Descriptive Statistics of Family-Level Variables								
	Mean	Std	Skew	P5	P25	Median	P75	P95
NFUNDS	4.709	0.406	-0.880	3.823	4.552	4.748	5.019	5.207
FAMSIZE (in \$ millions)	5,801	1,666	0.734	3,559	4,552	5,714	7,040	8,242
FAMOFFSET	0.015	0.004	-1.610	0.008	0.014	0.016	0.018	0.020
FAMOFFSETILLIQ	0.009	0.003	-0.662	0.006	0.007	0.009	0.010	0.013
FAMCOMHOLD	0.354	0.044	-0.505	0.271	0.338	0.354	0.391	0.415
FAMFLOWCORR	0.153	0.020	-0.817	0.122	0.145	0.155	0.168	0.174
FAMDIVERS	-0.689	0.124	0.951	-0.856	-0.763	-0.719	-0.655	-0.437
FAMFEEDISP (in %)	0.241	0.025	-1.170	0.208	0.225	0.245	0.259	0.274
FAMBENCHILLIQ (in %)	0.365	0.160	0.926	0.205	0.225	0.308	0.451	0.700
FAMCOMHOLDLIQ	0.205	0.040	-0.298	0.136	0.168	0.208	0.234	0.269
FAMCOMHOLDILLIQ	0.069	0.005	0.005	0.061	0.065	0.070	0.073	0.076
PANEL B: Descriptive Statistics of Fund-Level Variables								
	Mean	Std	Skew	P5	P25	Median	P75	P95
FUNDILLIQDEV (in %)	-0.022	0.036	-1.929	-0.103	-0.025	-0.014	0.002	0.010
FUNDEXPRATIO	0.013	0.001	-0.438	0.012	0.012	0.013	0.014	0.014
FUNDAGE (in years)	11.080	1.672	0.318	9.448	9.787	10.674	12.409	14.437
FUNDSIZE (in \$ millions)	1,272	386	2.436	815	1,013	1,223	1,441	1,682
FUNDRETVOL	0.051	0.019	0.218	0.027	0.032	0.052	0.069	0.077

Table 2: Continued

PANEL C: Descriptive Statistics of Family-Level Variables for Placebo Families									Diff. Between Actual and Placebo	
	Mean	Std	Skew	P5	P25	Median	P75	P95	Diff. in Means	p-value
NFUNDS	4.765	0.479	-0.910	3.894	4.509	4.826	5.151	5.342	-0.056	0.008
FAMSIZE	6.154	1.970	1.465	3,850	4,678	6,087	6,884	9,182	-353	0.009
FAMOFFSET	0.014	0.004	-1.027	0.006	0.012	0.015	0.017	0.020	0.001	0.050
FAMOFFSETILLIQ	0.007	0.002	-0.511	0.003	0.005	0.007	0.009	0.010	0.002	0.000
FAMCOMHOLD	0.155	0.022	-0.361	0.115	0.139	0.157	0.170	0.183	0.192	0.000
FAMFLOWCORR	0.054	0.049	4.264	0.009	0.026	0.046	0.067	0.088	0.099	0.000
FAMDIVERS	-0.690	0.125	0.979	-0.858	-0.766	-0.722	-0.653	-0.442	0.002	0.238
FAMFEEDISP	0.331	0.038	-0.907	0.269	0.309	0.342	0.363	0.375	-0.090	0.000

Table 2: Continued

PANEL D: Correlations of Family-Level Variables									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
NFUNDS	(1)	0.6296	0.4858	0.3515	0.6794	-0.1103	0.2648	0.2074	-0.0801
FAMSIZE	(2)	<.0001	0.2722	0.2004	0.5298	-0.0040	0.1169	0.0474	-0.0521
FAMOFFSET	(3)	<.0001	<.0001	0.8354	0.3616	-0.0593	0.0774	0.0944	-0.0723
FAMOFFSETILLIQ	(4)	<.0001	<.0001	<.0001	0.2469	-0.0372	0.0959	0.0482	0.0227
FAMCOMHOLD	(5)	<.0001	<.0001	<.0001	<.0001	-0.0117	0.1739	0.1684	-0.0749
FAMFLOWCORR	(6)	<.0001	<.0001	0.7320	0.3118	<.0001	-0.0125	-0.1128	0.0224
FAMDIVERS	(7)	<.0001	<.0001	<.0001	<.0001	0.2890	<.0001	0.0901	0.3558
FAMFEEDISP	(8)	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	-0.0400
FAMBENCHILLIQ	(9)	<.0001	<.0001	0.0607	<.0001	0.0537	<.0001	0.0004	

PANEL E: Correlations of Fund-Level Variables					
	(1)	(2)	(3)	(4)	(5)
FUNDILLIQDEV	(1)	0.0008	0.0243	0.0108	0.0318
FUNDEXPRATIO	(2)	0.8747	-0.1424	-0.1335	0.1062
FUNDAGE	(3)	<.0001	<.0001	0.2642	-0.0188
FUND SIZE	(4)	0.0416	<.0001	<.0001	-0.0423
FUNDRETVOL	(5)	<.0001	<.0001	0.0004	<.0001

Table 3: Style Diversity and Offsetting Trades in Mutual Fund Families

This table provides coefficient estimates obtained from Fama and MacBeth (1973) cross-sectional regressions. In Panel A, the dependent variable is FAMOFFSET (the minimum between buy and sell trades of a particular stock, aggregated across all the funds in a family, and then averaged across all the traded stocks in a given family-quarter). In Panel B, the dependent variable FAMOFFSETILLIQ only captures offsetting trades of illiquid stocks (i.e., stocks not in the bottom quintile of liquidity cost in the cross-section of stocks held by mutual funds in a given quarter). The control variables in both panels include our main independent variable of interest FAMDIVERS (the intra-family style diversity measure), FAMCOMHOLD (the extent of commonality in holdings across the funds within the family), FAMFLOWCORR (the average pairwise correlation of monthly flows across all the funds within the family), and FAMFEEDISP (the cross-sectional standard deviation of expense ratios across the funds within the family). For each family in our sample we create a 'placebo family' (column (10)) by randomly pairing each member fund with a fund that follows the same benchmark but does not belong in the same family. The standard errors (in parenthesis) are computed using the Newey and West (1987) correction. We denote by ***, **, * the significance at the 1%, 5%, and 10% levels, respectively.

PANEL A: Determinants of Offsetting Buy and Sell Trades in Fund Families									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	[Placebo Families] (10)
Dependent Variable: FAMOFFSET									
INTERCEPT	0.0552*** (0.0092)	0.0052*** (0.0003)	0.0160*** (0.0010)	0.0112*** (0.0008)	0.0257*** (0.0046)	0.0551*** (0.0090)	0.0508*** (0.0093)	0.0051*** (0.0006)	0.0020 (0.0028)
FAMDIVERS	0.0544*** (0.0096)				0.0266*** (0.0051)	0.0531*** (0.0091)	0.0524*** (0.0096)		-0.0013 (0.0037)
FAMCOMHOLD		0.0282*** (0.0023)			0.0256*** (0.0023)			0.0277*** (0.0023)	0.0715*** (0.0061)
FAMFLOWCORR			-0.0071*** (0.0020)		-0.0063*** (0.0016)			-0.0063*** (0.0016)	-0.0005 (0.0012)
FAMFEEDISP				0.0152*** (0.0019)			0.0120*** (0.0019)	0.0050** (0.0020)	-0.0008 (0.0010)
Observations	7,534	7,768	7,428	7,768	7,534	7,195	7,534	7,428	6,586
Adjusted R^2	0.0391*** (0.0075)	0.1289*** (0.0241)	0.0027 (0.0026)	0.0067*** (0.0018)	0.1359*** (0.0251)	0.0401*** (0.0088)	0.0428*** (0.0071)	0.1292*** (0.0251)	0.2725*** (0.0314)

Table 3: Continued

PANEL B: Determinants of Offsetting Buy and Sell Trades of Illiquid Assets in Fund Families										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	[Placebo Families] (10)
	Dependent Variable: FAMOFFSETILLIQ									
INTERCEPT	0.0324*** (0.0044)	0.0042*** (0.0004)	0.0097*** (0.0006)	0.0077*** (0.0006)	0.0183*** (0.0027)	0.0322*** (0.0042)	0.0308*** (0.0045)	0.0045*** (0.0007)	0.0184*** (0.0027)	0.0045*** (0.0017)
FAMDIVERS	0.0317*** (0.0047)				0.0185*** (0.0032)	0.0310*** (0.0044)	0.0311*** (0.0047)		0.0182*** (0.0032)	0.0035 (0.0021)
FAMCOMHOLD		0.0143*** (0.0009)			0.0123*** (0.0011)			0.0142*** (0.0010)	0.0123*** (0.0012)	0.0313*** (0.0031)
FAMFLOWCORR			-0.0033** (0.0012)			-0.0027*** (0.0009)		-0.0032*** (0.0010)	-0.0030*** (0.0009)	0.0000 (0.0008)
FAMFEEDISP				0.0062*** (0.0010)			0.0045*** (0.0011)	0.0006 (0.0012)	0.0005 (0.0012)	-0.0008 (0.0006)
Observations	6,636	6,856	6,567	6,856	6,636	6,348	6,636	6,567	6,348	6,438
Adjusted R^2	0.0388*** (0.0086)	0.0941*** (0.0187)	-0.0001 (0.0018)	0.0015 (0.0014)	0.1050*** (0.0202)	0.0378*** (0.0090)	0.0379*** (0.0088)	0.0895*** (0.0193)	0.1004*** (0.0207)	0.1805*** (0.0289)

Table 4: Style Diversity and Deviations from Benchmark Illiquidity

This table provides coefficient estimates obtained from Fama and MacBeth (1973) cross-sectional regressions. The dependent variable is FUNDILLIQDEV, the difference between the illiquidity of a fund and the illiquidity of its benchmark. In Panel A, the main independent variables of interest are the log of NFUNDS (the number of funds affiliated with a given family), the log of FAMSIZE (the fund family's AUM), and the intra-family style diversity measure FAMDIVERS (-1 times the minimum of the pairwise correlations of returns across all the benchmarks in a given family). We include some extra indicators. The indicator LOWBILLIQ (HIGHBILLIQ) equals one if the fund benchmark ranks in the bottom (top) tercile of illiquidity across all benchmarks in a given quarter. We also include the interaction terms between these indicators and each of the proxies for style diversity, i.e., LOGNFUNDS, LOGFAMSIZE, and FAMDIVERS. In Panel B we test for the competing hypothesis of cross-fund subsidization (Gaspar, Massa, and Matos (2006)). Therefore, we include two additional indicators (LOWFEE and HIGHFEE), their multiple interactions with the indicators LOWBILLIQ and HIGHBILLIQ, as well as our proxy for style diversity within a fund family (FAMDIVERS). The indicator LOWFEE (HIGHFEE) equals one if the fees charged by the funds rank in the bottom (top) tercile of fees charged across all the funds affiliated with the same family in a given quarter. The remaining covariates in both panels are the log of FUNDAGE (the number of years since the inception of the fund), the log of FUNDSIZE (the fund's TNA), as well as DG (an indicator for the investment style "Growth"), DGI (an indicator for the investment style "Growth and Income"), FUNDRETVOL (the monthly return volatility of the fund for the past 12 months), FUNDEXPRATIO (the fund's expense ratio), DLOSER (an indicator that equals one if the fund ranked in the bottom half of performance across all the funds in the same investment style), and FAMBENCHILLIQ (the average illiquidity of the benchmarks within the family). The standard errors (in parenthesis) are computed using the Newey and West (1987) correction. We denote by ***, **, * the significance at the 1%, 5%, and 10% levels, respectively.

PANEL A: Deviation of Fund Illiquidity from Benchmark Illiquidity				
Dependent Variable:	FUNDILLIQDEV			
	(1)	(2)	(3)	(4)
INTERCEPT	0.0507*** (0.0170)	0.0541*** (0.0175)	0.0715** (0.0297)	0.1530*** (0.0432)
LOWBILLIQ	0.0420*** (0.0106)	0.0340*** (0.0114)	0.1115*** (0.0177)	0.0877*** (0.0185)
HIGHBILLIQ	-0.1585*** (0.0373)	-0.1324*** (0.0244)	-0.3511*** (0.0645)	-0.4733*** (0.1049)
LOGNFUNDS	-0.0013 (0.0020)			-0.0062 (0.0077)
LOGNFUNDS $\times$ LOWBILLIQ	0.0046 (0.0046)			-0.0033 (0.0033)
LOGNFUNDS $\times$ HIGHBILLIQ	0.0122 (0.0218)			0.0546* (0.0289)
LOGFAMSIZE		-0.0035** (0.0013)		-0.0048** (0.0021)
LOGFAMSIZE $\times$ LOWBILLIQ		0.0023* (0.0012)		0.0009 (0.0013)
LOGFAMSIZE $\times$ HIGHBILLIQ		-0.0027 (0.0017)		-0.0037 (0.0025)
FAMDIVERS			0.0368 (0.0225)	0.1049*** (0.0338)
FAMDIVERS $\times$ LOWBILLIQ			0.0892*** (0.0171)	0.0590*** (0.0181)
FAMDIVERS $\times$ HIGHBILLIQ			-0.3188*** (0.0837)	-0.4336*** (0.1172)
Controls	Yes	Yes	Yes	Yes
Observations	33,254	33,254	33,254	33,254
Adjusted R^2	0.4768*** (0.0244)	0.4672*** (0.0213)	0.4897*** (0.0221)	0.4991*** (0.0235)

Table 4: Continued

PANEL B: Alternative Hypothesis of Cross-Subsidization		
Dependent Variable:	FUNDILLIQDEV	
	(1)	(2)
INTERCEPT	0.0389** (0.0183)	0.0046 (0.0897)
LOWBILLIQ	0.0502*** (0.0083)	0.1706** (0.0639)
HIGHBILLIQ	-0.1450*** (0.0190)	-0.1959* (0.1042)
LOWFEE	0.0004 (0.0035)	0.1505 (0.1510)
HIGHFEE	0.0007 (0.0050)	0.0793 (0.0762)
LOWFEE $\times$ LOWBILLIQ	0.004 (0.0047)	-0.055 (0.0425)
LOWFEE $\times$ HIGHBILLIQ	-0.0169 (0.0130)	-0.2395 (0.2087)
HIGHFEE $\times$ LOWBILLIQ	0.0084 (0.0082)	-0.0188 (0.0218)
HIGHFEE $\times$ HIGHBILLIQ	-0.0149* (0.0078)	-0.2519 (0.2024)
FAMDIVERS		-0.0676 (0.1164)
FAMDIVERS $\times$ LOWBILLIQ		0.1702** (0.0775)
FAMDIVERS $\times$ HIGHBILLIQ		-0.1224 (0.1449)
FAMDIVERS $\times$ LOWFEE		0.211 (0.2066)
FAMDIVERS $\times$ HIGHFEE		0.1081 (0.1070)
FAMDIVERS $\times$ LOWFEE $\times$ LOWBILLIQ		-0.0172 (0.0114)
FAMDIVERS $\times$ LOWFEE $\times$ HIGHBILLIQ		-0.3297 (0.2890)
FAMDIVERS $\times$ HIGHFEE $\times$ LOWBILLIQ		-0.0299 (0.0304)
FAMDIVERS $\times$ HIGHFEE $\times$ HIGHBILLIQ		-0.3558 (0.2892)
Controls	Yes	Yes
Observations	33,254	33,254
Adjusted R^2	0.4726*** (0.0229)	0.4920*** (0.0229)

Table 5: Implications for Fund Performance

This table provides coefficient estimates obtained from Fama and MacBeth (1973) cross-sectional regressions. The dependent variable is FRETADJ(q+1), the benchmark-adjusted performance of the fund at quarter q+1. In Columns (1)-(2) the regressions are performed for a subsample of passive funds, defined as the low tercile of active share in the cross-section. Columns (3)-(4) perform the regressions for a subsample of active funds, defined as the top tercile of active share in the cross-section. The main independent variable of interest is the intra-family style diversity measure FAMDIVERS(q) (-1 times the minimum of the pairwise correlations of returns across all the benchmarks in a given family), which is measured at quarter q. The additional indicators LOWBILLIQ and HIGHBILLIQ equal one if the fund benchmark ranks respectively in the bottom and top tercile of illiquidity across all benchmarks in a given quarter. We also interact these indicators with FAMDIVERS(q). The remaining covariates in both panels are the log of FUNDAGE (the number of years since the inception of the fund), the log of FUNDSIZE (the fund's TNA), as well as DG (an indicator for the investment style "Growth"), DGI (an indicator for the investment style "Growth and Income"), FUNDRETVOL (the monthly return volatility of the fund for the past 12 months), FUNDEXPRATIO (the fund's expense ratio), DLOSER (an indicator that equals one if the fund ranked in the bottom half of performance across all the funds in the same investment style), and FAMBENCHILLIQ (the average illiquidity of the benchmarks within the family). These controls are all measured at quarter q. The standard errors (in parenthesis) are computed using the Newey and West (1987) correction. We denote by ***, **, * the significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable:	FRETADJ(q+1)			
	Passive Funds		Active Funds	
	(1)	(2)	(3)	(4)
INTERCEPT	-0.1307 (0.1457)	-0.0649 (0.0874)	-0.0189 (0.0223)	-0.0097 (0.0348)
FAMDIVERS(q)	0.0573* (0.0334)	0.0719** (0.0320)	0.0083 (0.0204)	0.0198 (0.0183)
LOWBILLIQ		-0.0187 (0.0237)		0.0521* (0.0301)
HIGHBILLIQ		-0.0257 (0.0313)		-0.0565** (0.0259)
FAMDIVERS × LOWBILLIQ		-0.0524* (0.0301)		0.0342 (0.0219)
FAMDIVERS × HIGHBILLIQ		-0.0249 (0.0423)		-0.0772*** (0.0270)
Controls	Yes	Yes	Yes	Yes
Observations	10,738	10,738	11,351	11,351
Adjusted R^2	0.1459*** (0.0165)	0.3156*** (0.0226)	0.1012*** (0.0142)	0.1773*** (0.0216)

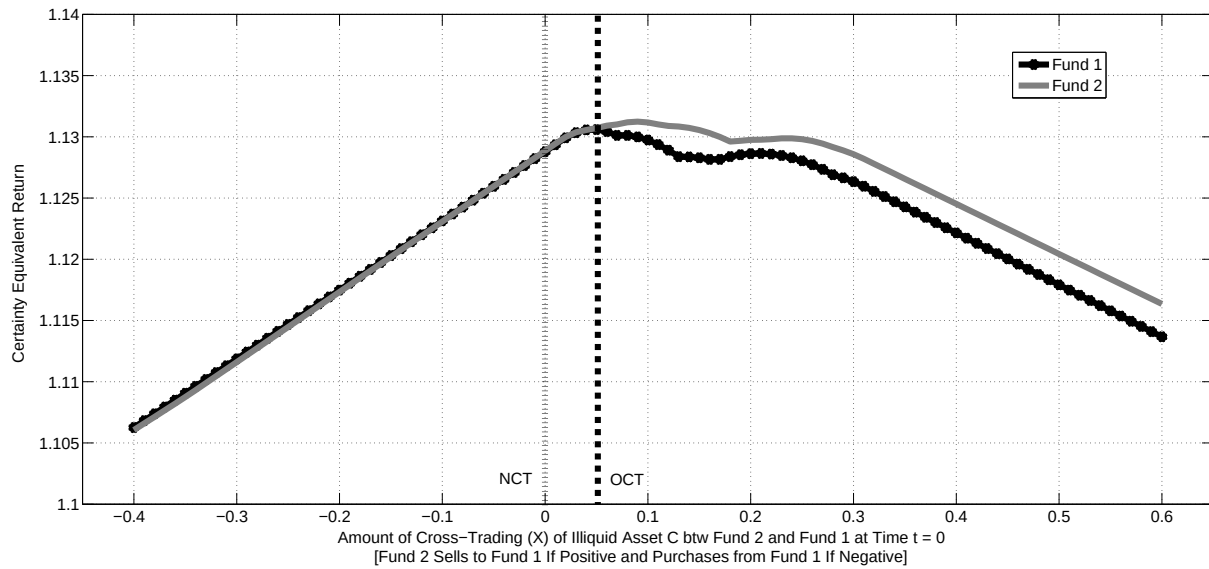


Figure 1: Optimal Cross-Trading

This plots the certainty equivalent (CE) rate of return each fund obtains for different levels of cross-trading. The vertical dotted line labelled as NCT represents the case in which both funds start off the investment period with portfolio compositions that exactly match those of their respective benchmarks. The vertical dashed line labelled as OCT identifies the level of cross-trading ($X = 0.05$) that improves the CE for both funds relative to NCT (i.e., $X = 0$). The funds follow the constrained max-min rule proposed in (12) to agree on the optimal level of cross-trading (OCT).

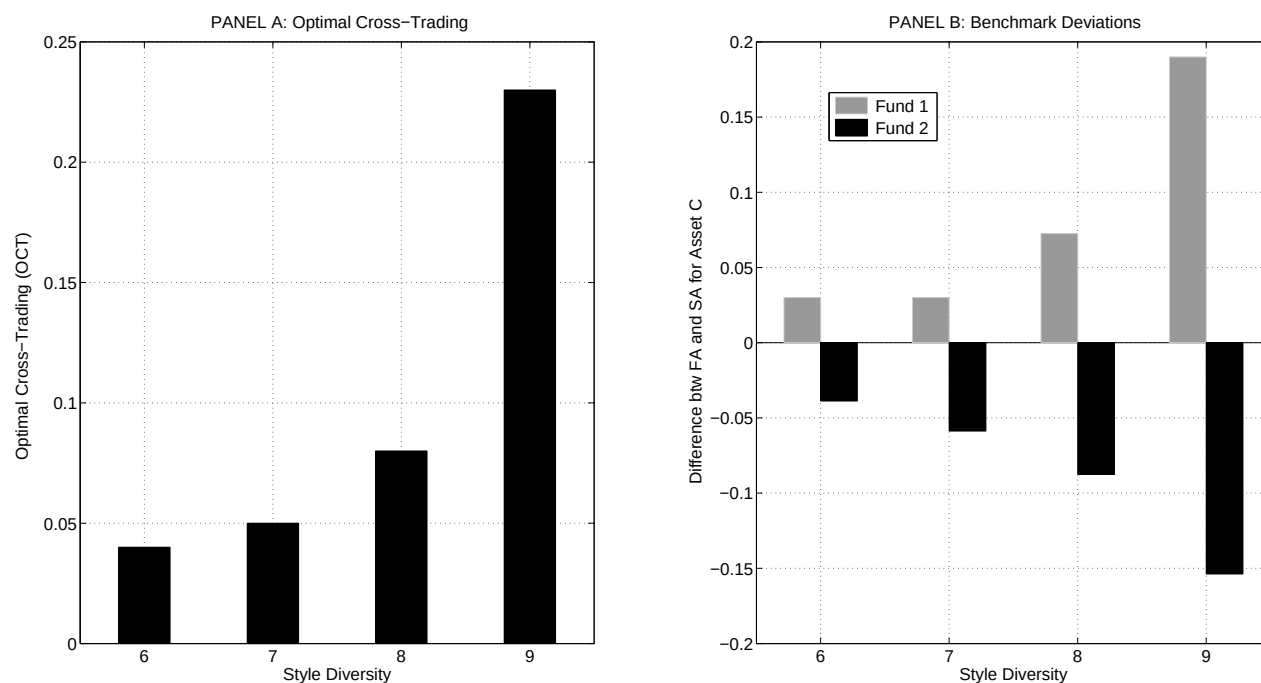


Figure 2: Style Diversity, Optimal Crossing, and Benchmark Deviations

Panel A reports the optimal level of cross-trading for different degrees of diversity within the fund family. Panel B plots the average excess holdings of illiquid asset C (comparing family-affiliated funds with their respective standalone versions) as a function of style diversity within the fund family. The style diversity in the x-axis is determined by the risk aversion parameter of household $h1$ (γ_{h1}).

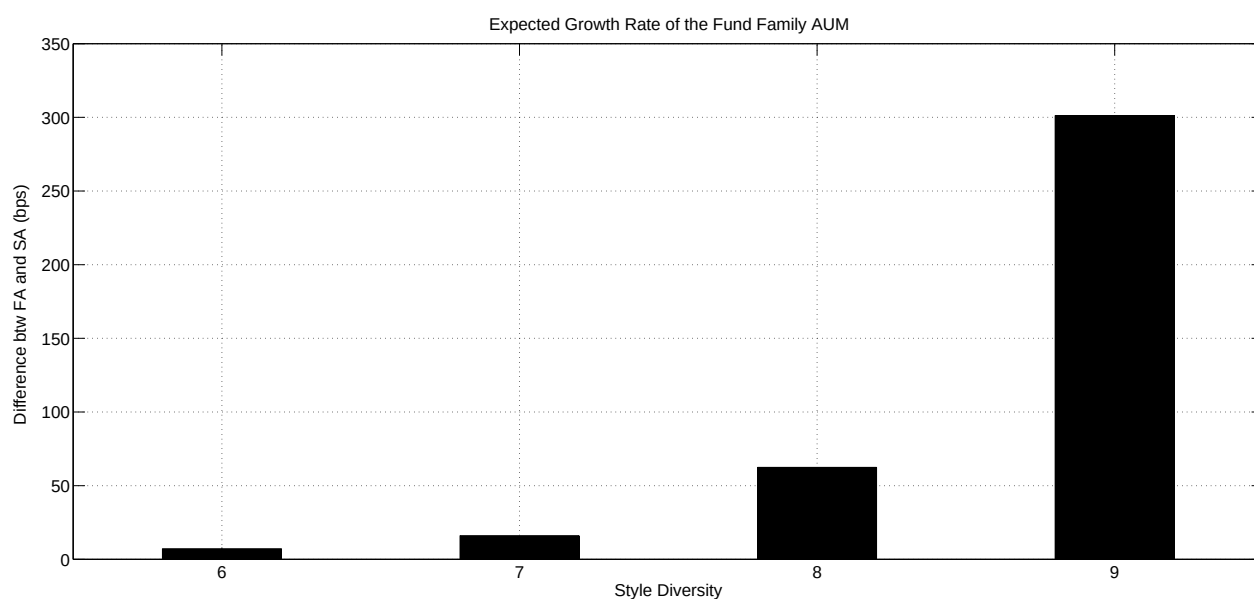


Figure 3: Expected Growth Rate of the Fund Family AUM

This figure plots the after-flow returns to the family (i.e., the expected growth rate of the assets under management) as a function of style diversity. The style diversity in the x-axis is determined by the risk aversion parameter of household $h1$ (γ_{h1}).

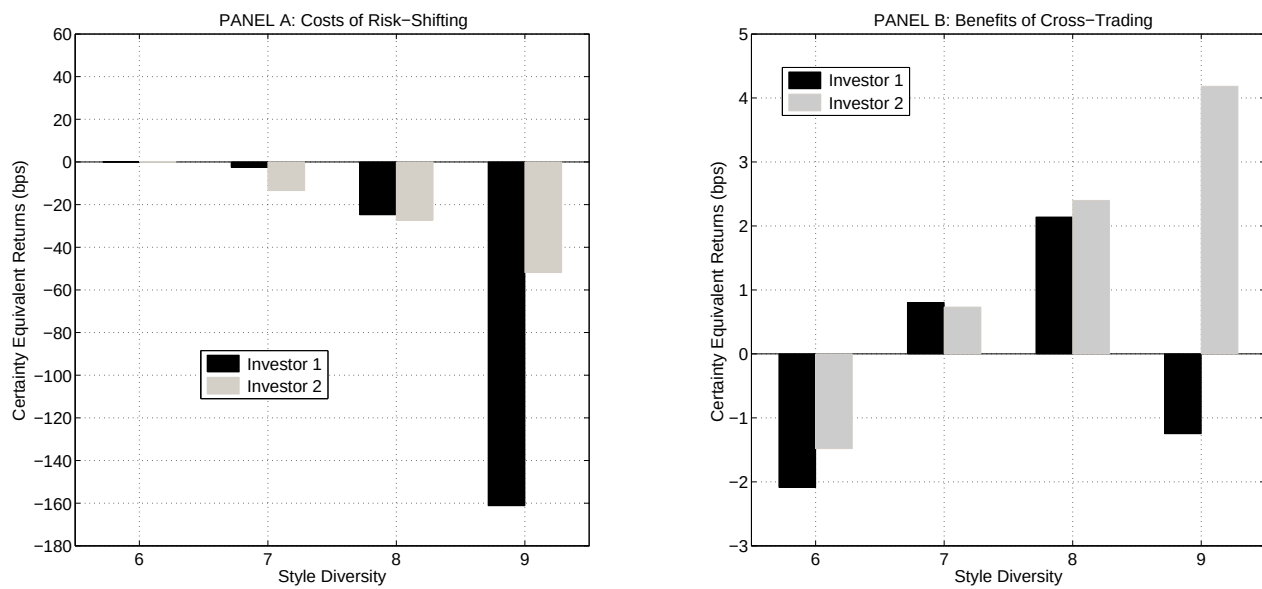


Figure 4: Costs and Benefits of Cross-Trading

This figure plots the net effect of risk-shifting for the households (Panel A), and the net effect of cross-trading under no-convexity kink (Panel B), as a function of style diversity. The style diversity in the x-axis is determined by the risk aversion parameter of household $h1$ (γ_{h1}).

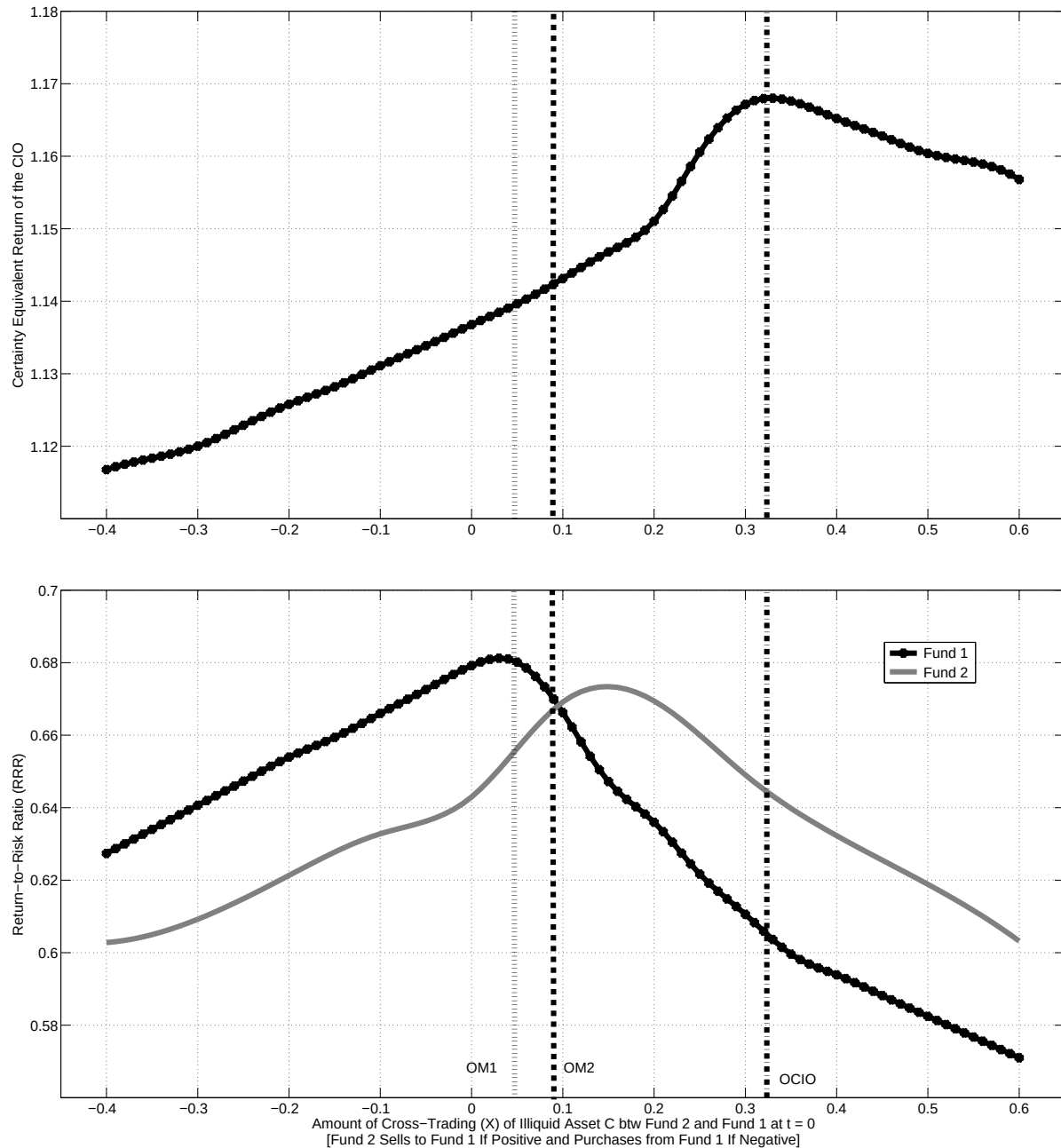


Figure 5: CER of the CIO and RRR of the Individual Funds

The top graph plots the certainty-equivalent (CE) rate of return that accrues to the CIO, and the bottom graph plots the return-to-risk ratio (RRR) of the individual funds, for different levels of cross-trading (X) between the two funds. The vertical dotted line labelled as *OM1* identifies the amount of cross-trading between the two funds that would maximize the utility of fund manager 1, while the vertical line labelled as *OM2* indicates the amount of cross-trading that would maximize the utility of fund manager 2. The vertical line labelled as *OCIO* indicates the amount of cross-trading that would maximize the utility of the CIO.

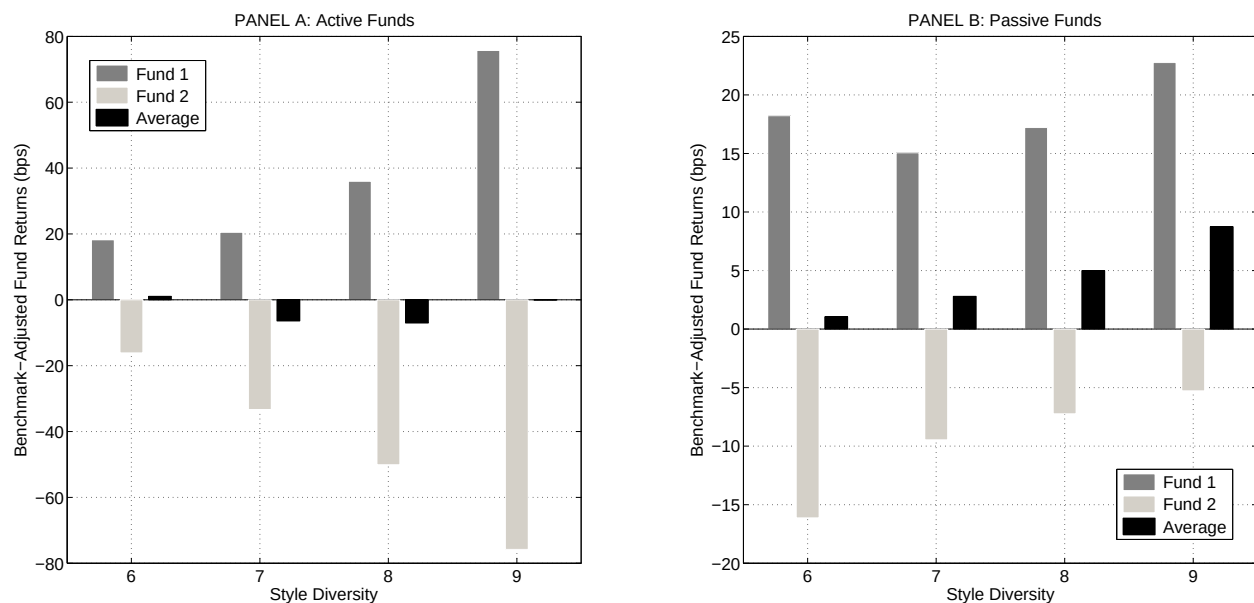


Figure 6: Benchmark-Adjusted Fund Returns

This figure plots benchmark-adjusted returns for both funds, as well as their average. Panel A shows the results for active funds, defined as funds with convex incentives. Panel B reports the results for passive funds, defined as funds with benchmark concerns but no convex incentives. The adjusted returns are plotted against style diversity. The style diversity in the x-axis is determined by the risk aversion parameter of household $h1$ (γ_{h1}).